Collision Ideals and Off-Diagonal Sheets

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Abstract

We study the scheme-theoretic collision relation of a polynomial self-map, namely its fiber product over the target, and describe the geometry away from the diagonal through boundary-supported inertia and Galois-closure sheets. For a polynomial map $F = (F_1, \ldots, F_n)\colon \mathbb{A}^n_{\mathbb{C}} \to \mathbb{A}^n_{\mathbb{C}}$, define in $S = \mathbb{C}[x_1, \ldots, x_n, y_1, \ldots, y_n]$ the collision and diagonal ideals

$$I_R = \big(F_i(x) - F_i(y)\big)_{i=1}^n, \qquad I_\Delta = (x_i - y_i)_{i=1}^n.$$

The image of I_Δ in the collision algebra $C_F := S/I_R$ is the obstruction ideal $\mathrm{Obs}(F) := I_\Delta/I_R$; it records the failure of the collision scheme $\operatorname{Spec} C_F$ to coincide with the diagonal. Its vanishing characterizes polynomial automorphisms:

$$\mathrm{Obs}(F) = 0 \iff F \text{ is a polynomial automorphism.}$$

In dimension two we give two module-theoretic formulations of the same fixed-map conclusion. Planar boundary coherence asks that the boundary local cohomology $H^1_{\partial X}(\overline{X}, \mathcal{O}_{\overline{X}})$ be a finite $\overline{A}$-module; it forces $\partial X = \varnothing$, hence makes the Keller map finite étale and therefore an automorphism. Planar ramification rigidity is the divisorial criterion that $\Omega_{A_N/B}$ have no height-one support, equivalently that it have finite length. On the common Galois normalization $Z = \operatorname{Spec} A_N$, we construct the pulled-back affine opens U_g, their canonical boundary ideals $\mathfrak{j}_g = I_{A_N}(Z \setminus U_g)$, and the fixed-locus ideals $\mathfrak{a}_C$. The height-one dictionary identifies moving-sheet coverage and finite-group boundary separation as exact criteria for ramification rigidity, after which purity and finite-étale rigidity give $N = L = K$ and collision vanishing. Although $\overline{A}$ and A_N are already finite over B, neither of these additional finiteness criteria follows formally. Universal validity of either criterion is equivalent to the two-dimensional Jacobian conjecture and is not proved here. The boundary divisor-class argument does, however, prove the normal/Galois case and therefore excludes generic degree two. For the separable nonnormal cubic class in dimension three, let

$$K = \mathbb{C}(F_1, F_2, F_3), \qquad L = \mathbb{C}(x_1, x_2, x_3),$$

and let N be the normal closure of L/K. Then

$$L \otimes_K L \simeq L \times N, \qquad \mathrm{Gal}(N/K) \simeq S_3.$$

The generic off-diagonal factor N is the ordered-root Galois-closure sheet. Consequently, $\mathrm{Obs}(F) \neq 0$. The algebraic construction, the pulled-back boundary dictionary, and the implications from supplied coverage or separation hypotheses are formalized in Lean 4. The boundary-coherence and finite-length support arguments remain explicit Lean interfaces whose manuscript proofs have not yet been formalized, as do the standard external geometric statements.

Contents

1 Introduction

The Jacobian conjecture asks whether a polynomial self-map of affine space with constant nonzero Jacobian determinant must be a polynomial automorphism. Dimension two remains open. Rather than treating injectivity as an external property of the map, this paper studies the scheme-theoretic fiber product of the map with itself and asks when its off-diagonal part disappears.

There is one divisorial endgame, expressed in several exact languages. The paper does not prove the Jacobian conjecture. It organizes the remaining problem around the following two module-theoretic formulations:

$$\begin{aligned}\mathrm{PBC}(F) &:\Longleftrightarrow H^1_{\partial X}(\overline{X}, \mathcal{O}_{\overline{X}}) \text{ is a finite } \overline{A}\text{-module},\\ \mathrm{PRR}(F) &:\Longleftrightarrow \Omega_{A_N/B} \text{ has no height-one support}\\ &\Longleftrightarrow \Omega_{A_N/B} \text{ has finite length over } A_N.\end{aligned}$$

Planar boundary coherence $\mathrm{PBC}(F)$ is the corresponding global properness formulation:

$$\mathrm{PBC}(F) \Longrightarrow \partial X = \varnothing \Longrightarrow f_F \text{ finite étale} \Longrightarrow F \text{ an automorphism.}$$

Planar ramification rigidity $\mathrm{PRR}(F)$ is the differential-module formulation of the codimension-one premise:

$$\mathrm{PRR}(F) \Longrightarrow Z/Y \text{ finite étale} \Longrightarrow N = K \Longrightarrow L = K \Longrightarrow q_F = 0 \Longrightarrow I_R = I_\Delta.$$

The divisorial endgame is controlled by an exact codimension-one obstruction on the Galois normalization. A ramified divisor E is eliminated as soon as one conjugate sheet moved by I_E still contains its generic point. In coordinates this is the conjugate-integrality condition

$$I_E \neq 1 \Longrightarrow \exists g \in G, \quad I_E \not\subseteq gHg^{-1}, \qquad w_E(g(x_1)), w_E(g(x_2)) \geq 0.$$

We call this planar moving-sheet coverage. Its ideal-theoretic form says that the fixed locus of a prime-order subgroup and the boundaries of all the sheets it moves have no common height-one component. Establishing this universally—equivalently, establishing planar ramification rigidity universally—is equivalent to $JC(2)$, so the reformulation locates the missing theorem without assuming it.

Only after all height-one points have been shown unramified do purity and finite-étale rigidity enter: they make the normal closure trivial and collapse the collision relation to the diagonal. This order is essential to the argument. The paper also develops boundary divisor-class and principal-parts criteria that may imply coverage under additional hypotheses; these are prospective mechanisms, not additional endgames or proofs of universal coverage. The divisor-class mechanism does prove unconditionally that a normal—hence Galois—marked extension L/K is trivial. In particular, a hypothetical planar counterexample must have L/K nonnormal and generic degree at least three.

The normalization rings $\overline{A}$ and A_N are finite B-modules from the outset. This makes $\overline{X}$ and Z affine varieties: they are integral affine schemes of finite type over $\mathbb{C}$. It does not make A finite over B, make boundary local cohomology finite, or make $\Omega_{A_N/B}$ finite length. Likewise, the vanishing of ordinary higher quasi-coherent cohomology on an affine scheme does not imply the vanishing or finiteness of local cohomology with support in its deleted boundary.

In the separable nonnormal cubic class in dimension three, the opposite phenomenon is visible generically: the off-diagonal factor is the ordered-root S_3-normal-closure sheet. This gives a structural comparison with the planar criteria while keeping its assumptions explicit.

Section 2 develops collision ideals and the dimension-independent normalization diagram. Section 3 constructs the planar pulled-back boundaries, proves the valuation dictionary, and states the equivalent coverage and separation criteria before applying the endgame. Section 4 treats the cubic S_3 class.

2 General construction

The organizing question is how the failure of a polynomial self-map to be one-to-one is encoded by the part of its fiber square lying off the diagonal. The collision ideal first isolates that part scheme-theoretically; the normalization diagram then describes it through conjugate sheets and inertia. These constructions are dimension-independent. Section 3 specializes them to planar Keller maps and locates the obstruction at the deleted normalization boundary, while Section 4 identifies the generic off-diagonal sheet in the separable nonnormal cubic class with an S_3-normal closure. The two specializations respectively describe when the collision scheme collapses to the diagonal and what its surviving generic sheet looks like when it does not.

2.1 The two ideals and their obstruction

Throughout the paper, the base field is $\mathbb{C}$. Let

$$F = (F_1, \dots, F_n)\colon X = \mathbb{A}^n_{\mathbb{C}} \longrightarrow \mathbb{A}^n_{\mathbb{C}}$$

be a polynomial map, and let

$$S = \mathbb{C}[x_1, \dots, x_n, y_1, \dots, y_n].$$

The map F is Keller if

$$\det JF \in \mathbb{C}^{\times}.$$

Define

$$I_R(F) = \big(F_i(x) - F_i(y)\big)_{i=1}^n, \qquad I_\Delta = (x_i - y_i)_{i=1}^n, \qquad \mathrm{Obs}(F) := I_\Delta / I_R(F).$$

The quotient $\mathrm{Obs}(F)$ is naturally an ideal of the collision ring $C_F := S/I_R(F)$; we call it the obstruction ideal. For a fixed map F, we abbreviate $I_R(F)$ to I_R.

Proposition 2.1 (Canonical containment)**.** *For every polynomial map F,*

$$I_R(F) \subseteq I_\Delta.$$

Proof. For $H \in \mathbb{C}[x_1, \dots, x_n]$, the difference $H(x) - H(y)$ belongs to $(x_1 - y_1, \dots, x_n - y_n)$. This follows by induction on H, or by telescoping each monomial one coordinate at a time. Apply the statement to each coordinate F_i. □

Let $A := \mathbb{C}[x_1, \dots, x_n]$. Diagonal evaluation induces a surjective ring map

$$\bar{\mu}_F\colon C_F \longrightarrow A, \qquad [s]_{I_R} \longmapsto s(x, x).$$

Proposition 2.2 (The diagonal kernel)**.** *As ideals of C_F,*

$$\ker(\bar{\mu}_F) = \mathrm{Obs}(F) = I_\Delta / I_R.$$

Consequently,

$$\ker(\bar{\mu}_F) = 0 \iff \mathrm{Obs}(F) = 0 \iff I_R = I_\Delta.$$

Proof. The kernel of diagonal evaluation $S \to A$ is I_Δ. Passing through S/I_R therefore identifies the displayed kernel with I_Δ / I_R. Since $I_R \subseteq I_\Delta$,

$$I_\Delta / I_R = 0 \iff I_R = I_\Delta.$$

□

2.2 The fiber-product interpretation

Let

$$A = \mathbb{C}[x_1, \dots, x_n], \qquad B = \mathbb{C}[F_1, \dots, F_n] \subseteq A.$$

Let $X = \operatorname{Spec} A$, let $Y = \operatorname{Spec} B$, and let

$$f_F \colon X \longrightarrow Y$$

be the morphism induced by $B \hookrightarrow A$. The homomorphisms $\iota_x, \iota_y \colon A \rightrightarrows C_F$, defined by $\iota_x(a) = [a(x)]$ and $\iota_y(a) = [a(y)]$, agree on B because $[F_i(x)] = [F_i(y)]$ in C_F. Their common restriction equips C_F with a B-algebra structure.

Theorem 2.3 (Fiber-product interpretation)**.** *There is a canonical B-algebra equivalence and its dual affine-scheme equivalence*

$$C_F \cong A \otimes_B A, \qquad R_F := \operatorname{Spec} C_F \cong X \times_Y X.$$

Under the first equivalence, $\bar{\mu}_F$ is tensor multiplication

$$A \otimes_B A \longrightarrow A, \qquad a \otimes a' \longmapsto aa'.$$

Its dual morphism is the diagonal

$$\Delta_{X/Y} \colon X \longrightarrow X \times_Y X.$$

Proof. A map $C_F \to T$ to a commutative $\mathbb{C}$-algebra T is equivalent to a pair of maps $f, g \colon A \rightrightarrows T$ satisfying

$$f(F_i) = g(F_i) \qquad (1 \le i \le n).$$

Since the F_i generate B, this is precisely the universal property of $A \otimes_B A$. Applying Spec to this equivalence and using

$$\operatorname{Spec}(A \otimes_B A) \cong \operatorname{Spec} A \times_{\operatorname{Spec} B} \operatorname{Spec} A$$

gives the asserted fiber-product equivalence. The statements about multiplication and the diagonal follow on pure tensors. $\square$

$$R_F(\mathbb{C}) = \{(u, v) \in X(\mathbb{C}) \times X(\mathbb{C}) : F(u) = F(v)\}.$$

Inside

$$X \times X = \operatorname{Spec} S,$$

the ideals I_R and I_Δ define, respectively, $X \times_Y X$ and the image of $\Delta_{X/Y}$. The contravariance of $I \mapsto V(I)$ therefore gives the equivalences

$$I_R \subseteq I_\Delta \quad \Longleftrightarrow \quad \Delta_{X/Y}(X) \subseteq X \times_Y X$$

and

$$I_R = I_\Delta \quad \Longleftrightarrow \quad X \times_Y X = \Delta_{X/Y}(X)$$

as closed subschemes of $X \times X$. Thus the algebraic and geometric gaps are two descriptions of the same scheme-theoretic containment.

2.3 The off-diagonal factor and secant projectors

For every polynomial map F, define

$$I_{\mathrm{off}} := I_R : I_\Delta, \qquad C_F^\circ := S/I_{\mathrm{off}}, \qquad R_F^\circ := \operatorname{Spec} C_F^\circ.$$

This definition does not require the diagonal to be clopen.

Corollary 2.4 (Off-diagonal emptiness criterion)**.** *For every polynomial map F, the following are equivalent:*

$$R_F^\circ = \varnothing \iff C_F^\circ = 0 \iff \operatorname{Obs}(F) = 0 \iff I_R = I_\Delta.$$

Proof.

$$R_F^\circ = \varnothing \iff C_F^\circ = 0 \iff I_R : I_\Delta = S.$$

Moreover,

$$I_R : I_\Delta = S \iff I_\Delta \subseteq I_R \iff I_R = I_\Delta.$$

The equivalence with $\operatorname{Obs}(F) = 0$ is Proposition 2.2. □

The idempotent mechanism behind a clopen diagonal is purely commutative-algebraic. Let C be a commutative ring, $J \subseteq C$ an ideal, $\delta \in C$, and $c \in C^\times$. Suppose

$$\delta J = 0, \qquad \delta - c \in J.$$

Define

$$q := 1 - c^{-1}\delta.$$

Theorem 2.5 (Abstract off-diagonal projector)**.** *The element q is idempotent and*

$$J = Cq, \qquad \operatorname{Ann}_C(J) = C(1-q).$$

In particular,

$$J = 0 \iff q = 0.$$

Proof. The congruence $\delta \equiv c \pmod{J}$ gives $q \in J$. For every $j \in J$,

$$qj = j - c^{-1}\delta j = j.$$

Taking $j = q$ proves $q^2 = q$; the same identity gives $J = Cq$. Finally,

$$aq = 0 \iff a = a(1-q),$$

so $\operatorname{Ann}_C(J) = \operatorname{Ann}_C(Cq) = C(1-q)$. □

Whenever the theorem is applied with

$$C = C_F, \qquad J = \operatorname{Obs}(F),$$

the identities $J = C_F q$ and $C_F/J \cong A$ give

$$C_F \cong A \times C_F/C_F(1-q).$$

Thus the first factor is the diagonal and the second is its off-diagonal complement.

2.4 Generic fibers and marked-root decomposition

Let F be dominant and generically finite, and let

$$A = \mathbb{C}[x_1, \dots, x_n], \qquad B = \mathbb{C}[F_1, \dots, F_n] \subseteq A.$$

Let

$$K = \mathrm{Frac}(B), \qquad L = \mathrm{Frac}(A).$$

Let

$$\theta \colon K \otimes_B A \longrightarrow L, \qquad \lambda \otimes a \longmapsto \lambda a.$$

Theorem 2.6 (Generic collision bridge)**.** *The map θ is an isomorphism. Consequently, there is a K-algebra isomorphism*

$$K \otimes_B C_F \cong L \otimes_K L,$$

and the following square commutes:

$$\begin{array}{ccc} K \otimes_B C_F & \xrightarrow{1\otimes\bar{\mu}_F} & K \otimes_B A \\ \sim\downarrow & & \downarrow\theta\ \sim \\ L \otimes_K L & \xrightarrow{\mu_L} & L. \end{array}$$

Proof. Let $U = B \setminus \{0\}$. Since $K = U^{-1}B$, localization gives

$$K \otimes_B A \xrightarrow{\sim} U^{-1}A, \qquad \frac{b}{u} \otimes a \longmapsto \frac{ba}{u}.$$

Since $A = \mathbb{C}[x_1, \dots, x_n]$ is a domain and $B \subseteq A$,

$$B \setminus \{0\} \subseteq A \setminus \{0\}.$$

Hence

$$U^{-1}A \hookrightarrow \mathrm{Frac}(A) = L, \qquad \frac{a}{u} \longmapsto \frac{a}{u}.$$

Composing this injection with the displayed isomorphism gives θ, so θ is injective.

Since F is generically finite, L/K is a finite extension. The image of $U^{-1}A$ in L is

$$K[x_1, \dots, x_n].$$

Since each x_i is algebraic over K, this K-domain is finite-dimensional over K, hence is a field. Since it contains A, it contains $\mathrm{Frac}(A) = L$; the reverse inclusion is already given by its embedding into L. Therefore

$$U^{-1}A = L,$$

and θ is an isomorphism.

By Theorem 2.3, identify C_F with $A \otimes_B A$. There is a K-algebra isomorphism

$$\beta \colon K \otimes_B (A \otimes_B A) \xrightarrow{\sim} (K \otimes_B A) \otimes_K (K \otimes_B A)$$

defined on pure tensors by

$$\beta(\lambda \otimes (a \otimes a')) = (\lambda \otimes a) \otimes (1 \otimes a').$$

Its inverse is

$$(\lambda \otimes a) \otimes (\lambda' \otimes a') \longmapsto \lambda\lambda' \otimes (a \otimes a').$$

The balancing relations over B and K show that these formulas are well-defined, and the two displayed maps are mutually inverse. Composing β with $\theta \otimes_K \theta$ gives

$$\Phi_F \colon K \otimes_B C_F \xrightarrow{\sim} L \otimes_K L.$$

It remains to identify the diagonal map. Under $C_F \cong A \otimes_B A$, the homomorphism $\bar{\mu}_F$ is multiplication. Therefore, on a pure tensor,

$$(\mu_L \circ \Phi_F)\big(\lambda \otimes (a \otimes a')\big) = (\lambda a)a' = \lambda a a',$$
$$\big(\theta \circ (1 \otimes \bar{\mu}_F)\big)\big(\lambda \otimes (a \otimes a')\big) = \theta(\lambda \otimes aa') = \lambda a a'.$$

Pure tensors generate $K \otimes_B C_F$, so the two homomorphisms are equal. This proves the commutative square. □

Suppose now that L/K is separable and has a power basis $L = K(\alpha)$. Let $m_\alpha(T) \in K[T]$ be the minimal polynomial of α, and let $h_\alpha(T) \in L[T]$ be determined by

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \qquad \text{in } L[T].$$

Theorem 2.7 (Marked-root tensor decomposition)**.** *Give $L \otimes_K L$ its L-algebra structure through the left tensor factor. There is an L-algebra equivalence*

$$L \otimes_K L \cong L \times L[T]/(h_\alpha),$$

under which tensor multiplication $L \otimes_K L \to L$ is projection onto the first factor.

Proof. The power-basis presentation

$$L \cong K[T]/(m_\alpha)$$

and base change from K to L give an L-algebra isomorphism

$$L \otimes_K L \xrightarrow{\sim} L[T]/(m_\alpha), \qquad 1 \otimes \alpha \longmapsto T.$$

Since m_α is separable, $m'_\alpha(\alpha) \neq 0$. Differentiating

$$m_\alpha(T) = (T - \alpha)h_\alpha(T)$$

and evaluating at $T = \alpha$ gives

$$h_\alpha(\alpha) = m'_\alpha(\alpha) \neq 0.$$

Thus $T - \alpha$ does not divide h_α, so these two polynomials are coprime in $L[T]$. The Chinese remainder theorem now gives

$$L[T]/(m_\alpha) \xrightarrow{\sim} L[T]/(T - \alpha) \times L[T]/(h_\alpha) \xrightarrow{\sim} L \times L[T]/(h_\alpha).$$

Under the first isomorphism above, a pure tensor $\ell \otimes p(\alpha)$, with $p \in K[T]$, corresponds to the class of $\ell p(T)$. Tensor multiplication sends it to $\ell p(\alpha)$, which is evaluation at $T = \alpha$. Under the Chinese remainder isomorphism, this evaluation map is projection onto the factor $L[T]/(T - \alpha) \cong L$. Hence tensor multiplication is the first projection. □

Together with Theorem 2.6, this gives

$$K \otimes_B C_F \cong L \times L[T]/(h_\alpha),$$

where the first projection is the generic diagonal.

2.5 Normalized covers and inertia

Let L/K be finite and separable, and let N/K be its normal closure, with a fixed embedding $L \hookrightarrow N$. Let

$$G = \mathrm{Gal}(N/K), \qquad H = \mathrm{Gal}(N/L).$$

Proposition 2.8 (Conjugate-embedding action)**.** *Restriction induces a bijection*

$$G/H \xrightarrow{\sim} \mathrm{Hom}_K(L, N), \qquad gH \longmapsto g|_L.$$

The induced action of G on G/H is faithful; equivalently,

$$\mathrm{core}_G(H) = \bigcap_{g \in G} gHg^{-1} = 1.$$

Proof. For $g_1, g_2 \in G$,

$$g_1|_L = g_2|_L \quad \Longleftrightarrow \quad g_2^{-1} g_1 \in H \quad \Longleftrightarrow \quad g_1 H = g_2 H.$$

Every K-embedding $L \hookrightarrow N$ extends to an element of G, so restriction gives the stated bijection. Its action kernel is

$$\ker(G \curvearrowright G/H) = \bigcap_{g \in G} gHg^{-1} = \mathrm{core}_G(H).$$

If $\sigma \in \mathrm{core}_G(H)$, then

$$\sigma|_{g(L)} = \mathrm{id}_{g(L)} \quad (g \in G).$$

Since N is the normal closure,

$$N = K\big(g(L) : g \in G\big),$$

and therefore $\sigma = \mathrm{id}_N$. $\square$

Let

$$Y = \mathrm{Spec}\, B, \qquad \overline{X} = \mathrm{Norm}_L(Y), \qquad Z = \mathrm{Norm}_N(Y).$$

The marked embedding $L \hookrightarrow N$ induces a canonical commutative triangle

$$Z \xrightarrow{\pi} \overline{X} \xrightarrow{\bar\nu} Y, \qquad \nu = \bar\nu \circ \pi \colon Z \longrightarrow Y.$$

Suppose that $\bar\nu$ and ν are finite, and fix an open immersion over Y

$$j \colon X = \mathrm{Spec}\, A \hookrightarrow \overline{X}$$

with $\bar\nu \circ j = f_F$. Denote this diagram and its deleted boundary by

$$\mathcal{D} = \Big(Z \xrightarrow{\pi} \overline{X} \xrightarrow{\bar\nu} Y,\, j \colon X \hookrightarrow \overline{X}\Big), \qquad \partial X = \overline{X} \setminus j(X).$$

Fix $E \in Z^{(1)}$. Let w_E be its divisorial valuation, let $b = \nu(E)$, and let $v_b = w_E|_K$. Let $D_E \leq G$ and $I_E \trianglelefteq D_E$ be its decomposition and inertia groups. Let

$$\mathcal{Q}_E := D_E \backslash G / H.$$

For $\omega = D_E g H \in \mathcal{Q}_E$, let

$$u_\omega = (g^{-1} w_E)|_L, \qquad \mathrm{cent}(E, \omega) = \mathrm{cent}_{\overline{X}}(u_\omega),$$

and let

$$i_E(\omega) = [I_E : I_E \cap gHg^{-1}].$$

Proposition 2.9 (Conjugate divisors and double cosets)**.** *The assignments above are well-defined, and restriction of valuations induces a bijection*

$$\mathcal{Q}_E \xrightarrow{\sim} \{\textit{prime divisors of } \overline{X} \textit{ above } b\}, \qquad \omega \longmapsto \mathrm{cent}(E, \omega).$$

Proof. The group G acts transitively on the extensions of v_b to N, and the stabilizer of w_E is D_E. Restricting such a valuation to $L = N^H$ identifies precisely those extensions whose indices differ on the right by an element of H. The resulting orbits are therefore indexed by $D_E \backslash G/H$, giving the stated bijection.

It remains to check independence of representatives. If $D_E g' H = D_E g H$, then $g' = dgh$ for some $d \in D_E$ and $h \in H$. The element d fixes w_E, while h fixes L pointwise, so

$$(g'^{-1} w_E)|_L = (g^{-1} w_E)|_L.$$

Moreover, $I_E \trianglelefteq D_E$, and conjugation by d identifies

$$I_E \cap g' H g'^{-1} \quad \text{with} \quad I_E \cap g H g^{-1}.$$

Thus u_ω, $\mathrm{cent}(E, \omega)$, and $i_E(\omega)$ are well-defined. □

We use the notation

$$e\big(\mathrm{cent}(E, \omega)/b\big) := e(u_\omega / v_b).$$

Since the residue characteristic is zero,

$$\mathrm{Ram}^{(1)}(Z/Y) := \{E \in Z^{(1)} : e(w_E/v_b) > 1\} = \{E \in Z^{(1)} : I_E \neq 1\}.$$

Proposition 2.10 (Detection of inertia on a conjugate sheet)**.** *If $I_E \neq 1$, then there is an $\omega \in \mathcal{Q}_E$ with*

$$i_E(\omega) > 1.$$

Proof. Suppose, to the contrary, that $i_E(\omega) = 1$ for every $\omega \in \mathcal{Q}_E$. For each $g \in G$,

$$1 = i_E(D_E g H) = [I_E : I_E \cap g H g^{-1}],$$

so

$$I_E = I_E \cap g H g^{-1} \subseteq g H g^{-1} \qquad \text{for every } g \in G.$$

Consequently,

$$I_E \subseteq \bigcap_{g \in G} g H g^{-1} = \mathrm{core}_G(H) = 1,$$

contrary to $I_E \neq 1$. □

Proposition 2.11 (Intermediate ramification index)**.** *For $\omega = D_E g H \in \mathcal{Q}_E$,*

$$e\big(\mathrm{cent}(E, \omega)/b\big) = i_E(\omega) = [I_E : I_E \cap g H g^{-1}].$$

Proof. Let w_E and v_b be the divisorial valuations associated with E and b, and let

$$w_g = g^{-1} w_E, \qquad u_\omega = w_g|_L.$$

The valuation u_ω has center $\mathrm{cent}(E,\omega)$ on $\overline{X}$. Since the residue fields have characteristic zero,

$$e(w_g/v_b) = |I_E|.$$

The inertia group of w_g over u_ω is

$$g^{-1}I_E g \cap H,$$

and hence

$$e(w_g/u_\omega) = |g^{-1}I_E g \cap H| = |I_E \cap gHg^{-1}|.$$

Multiplicativity of ramification indices in the field tower

$$K \subseteq L \subseteq N$$

therefore gives

$$e(u_\omega/v_b) = \frac{e(w_g/v_b)}{e(w_g/u_\omega)} = [I_E : I_E \cap gHg^{-1}],$$

which is the asserted formula. $\square$

Proposition 2.12 (Visibility-to-boundary)**.** *Suppose that $X \to Y$ is étale. For $E \in Z^{(1)}$ and $\omega \in \mathcal{Q}_E$,*

$$i_E(\omega) > 1 \quad \Longrightarrow \quad \mathrm{cent}(E,\omega) \in \partial X.$$

Proof. Suppose that $\mathrm{cent}(E,\omega) \in j(X)$, and let $x \in X$ be the corresponding point under the open immersion j. The local homomorphism

$$\mathcal{O}_{Y,b} \longrightarrow \mathcal{O}_{X,x}$$

is étale, so $e(\mathrm{cent}(E,\omega)/b) = 1$. Proposition 2.11 now gives

$$i_E(\omega) = e\big(\mathrm{cent}(E,\omega)/b\big) = 1.$$

Taking the contrapositive gives the result. $\square$

Definition 2.13 (Hidden-inertia locus)**.**

$$\mathcal{H}(\mathcal{D}) := \left\{ E \in \mathrm{Ram}^{(1)}(Z/Y) : \begin{array}{l} \exists \omega \in \mathcal{Q}_E,\ i_E(\omega) > 1, \\ \forall \omega \in \mathcal{Q}_E, \quad i_E(\omega) > 1 \Longrightarrow \mathrm{cent}(E,\omega) \in \partial X \end{array} \right\}.$$

Thus $E \in \mathcal{H}(\mathcal{D})$ precisely when nontrivial inertia is visible on a conjugate sheet but every visible center belongs to the deleted boundary.

Proposition 2.14 (Divisorial ramification is hidden from the étale sheet)**.** *Suppose that N/K is the normal closure of L/K, the residue characteristic is zero, and $X \to Y$ is étale. Then*

$$\mathcal{H}(\mathcal{D}) = \mathrm{Ram}^{(1)}(Z/Y).$$

Equivalently, every divisorial ramification orbit has a conjugate sheet on which its inertia acts nontrivially, but every such center lies in the deleted boundary.

Proof. The inclusion $\mathcal{H}(\mathcal{D}) \subseteq \mathrm{Ram}^{(1)}(Z/Y)$ is part of the definition. Conversely, let $E \in \mathrm{Ram}^{(1)}(Z/Y)$. In residue characteristic zero, $I_E \neq 1$. Since N is the normal closure of L/K, Proposition 2.8 gives $\mathrm{core}_G(H) = 1$, and Proposition 2.10 supplies an $\omega \in \mathcal{Q}_E$ with $i_E(\omega) > 1$. For every $\omega' \in \mathcal{Q}_E$, Proposition 2.12 gives

$$i_E(\omega') > 1 \quad \Longrightarrow \quad \mathrm{cent}(E,\omega') \in \partial X.$$

These are exactly the two conditions defining $E \in \mathcal{H}(\mathcal{D})$. $\square$

2.6 Automorphism detection

Theorem 2.15 (Automorphism criterion)**.** *For an arbitrary polynomial self-map $F\colon \mathbb{A}^n_{\mathbb{C}} \to \mathbb{A}^n_{\mathbb{C}}$, the following are equivalent:*

(1) *F is a polynomial automorphism;*

(2) *F is injective on $\mathbb{A}^n(\mathbb{C})$;*

(3) *$I_R(F) = I_\Delta$;*

(4) *$\mathrm{Obs}(F) = 0$.*

Proof. Suppose first that F has polynomial inverse $G = (G_1, \ldots, G_n)$. Then

$$x_i - y_i = G_i(F(x)) - G_i(F(y)) \in I_R$$

for every i, so $I_\Delta \subseteq I_R$. Proposition 2.1 gives equality. Thus (1) implies (3).

Suppose $I_R = I_\Delta$, and let $\mathrm{ev}_{(a,b)}\colon S \to \mathbb{C}$ be evaluation at (a, b). If $F(a) = F(b)$, then

$$I_R \subseteq \ker(\mathrm{ev}_{(a,b)}).$$

Therefore $x_i - y_i \in I_\Delta = I_R$ gives $a_i - b_i = 0$ for every i. Thus $a = b$, proving (3)⇒(2). The Ax–Grothendieck theorem [Ax69] gives (2)⇒(1). Finally, Proposition 2.2 gives (3)⟺(4). □

Corollary 2.16 (The Jacobian conjecture as obstruction vanishing)**.** *For every $n \geq 1$, the n-dimensional Jacobian conjecture is equivalent to each of the statements*

$$\forall F\colon \mathbb{A}^n_{\mathbb{C}} \to \mathbb{A}^n_{\mathbb{C}}, \qquad \det JF \in \mathbb{C}^\times \quad \Longrightarrow \quad \mathrm{Obs}(F) = 0,$$

and

$$\forall F\colon \mathbb{A}^n_{\mathbb{C}} \to \mathbb{A}^n_{\mathbb{C}}, \qquad \det JF \in \mathbb{C}^\times \quad \Longrightarrow \quad I_R(F) = I_\Delta.$$

Proof. Apply Theorem 2.15 to every Keller map. Its equivalent conditions (1), (3), and (4) give the two displayed formulations. □

2.7 Generic degree one

Corollary 2.17 (Generic degree-one descent)**.** *If*

$$L = K \qquad \text{and} \qquad A \text{ is flat over } B,$$

then

$$\mathrm{Obs}(F) = 0.$$

Proof. When $L = K$, Theorem 2.6 identifies the base-changed diagonal with tensor multiplication:

$$\ker(1 \otimes \bar{\mu}_F) \cong \ker\Big(K \otimes_K K \xrightarrow{\mu_K} K\Big) = 0.$$

The fiber-product identification of Theorem 2.3 is

$$C_F \xrightarrow{\sim} A \otimes_B A.$$

Since A is flat over B, the tensor product $A \otimes_B A$, and therefore C_F, is flat over B. The localization map

$$\iota_C \colon C_F \hookrightarrow K \otimes_B C_F$$

is therefore injective, and

$$\iota_C(\ker \bar{\mu}_F) \subseteq \ker(1 \otimes \bar{\mu}_F) = 0.$$

Thus $\ker \bar{\mu}_F = 0$. Proposition 2.2 identifies

$$\ker(\bar{\mu}_F) = \mathrm{Obs}(F) = I_\Delta / I_R,$$

so $\mathrm{Obs}(F) = 0$. $\square$

Lean correspondence. `CollisionIdeals/CollisionDiagonal.lean`, `CollisionIdeals/OffDiagonalScheme.lean`, `CollisionIdeals/Secant.lean`, `CollisionIdeals/GenericCollisionDiagonal.lean`, `CollisionIdeals/KellerCollisionModel.lean`, and `CollisionIdeals/JacobianConjecture.lean`. The Lean form of Theorem 2.6 takes surjectivity of θ as a parameter; the proof above derives it from generic finiteness.

3 Planar vanishing

We now specialize the dimension-independent collision, normalization, and inertia constructions of Section 2 to dimension two. The purpose of this section is to identify a concrete boundary obstruction, not to assume it away implicitly: the secant calculation separates the diagonal from the off-diagonal collision sheet, while the normalization calculation shows exactly how nontrivial inertia would have to escape every relevant conjugate affine sheet.

Let

$$F = (P, Q) \colon \mathbb{A}^2_{\mathbb{C}} \longrightarrow \mathbb{A}^2_{\mathbb{C}}$$

be a polynomial map satisfying $\det JF = c \in \mathbb{C}^\times$.

Set, from this point onward,

$$A = \mathbb{C}[x_1, x_2], \qquad B = \mathbb{C}[P, Q] \subseteq A, \qquad K = \mathbb{C}(P, Q), \qquad L = \mathbb{C}(x_1, x_2),$$

and let $X = \operatorname{Spec} A$, $Y = \operatorname{Spec} B$.

Proposition 3.1 (Planar Keller geometry)**.**

$$\mathbb{C}[u, v] \xrightarrow{\sim} B, \qquad u \longmapsto P, \quad v \longmapsto Q.$$

Moreover,

$$X \xrightarrow{f_F} Y \text{ is étale}, \qquad A \text{ is flat over } B, \qquad [L : K] < \infty \text{ and } L/K \text{ is separable.}$$

Proof. The identity

$$dP \wedge dQ = \left(\frac{\partial P}{\partial x_1} \frac{\partial Q}{\partial x_2} - \frac{\partial P}{\partial x_2} \frac{\partial Q}{\partial x_1} \right) dx_1 \wedge dx_2 = c \, dx_1 \wedge dx_2 \neq 0$$

shows that P, Q are algebraically independent. Thus $B \cong \mathbb{C}[u, v]$ and $Y \cong \mathbb{A}^2_{\mathbb{C}}$. Moreover,

$$A \cong \mathbb{C}[u, v, U, V] / \big(P(U, V) - u,\ Q(U, V) - v\big)$$

as a $\mathbb{C}[u, v]$-algebra, and its relative Jacobian determinant is $c \in A^\times$. The Jacobian criterion therefore makes $B \to A$ étale. In particular it is flat and quasi-finite. Since f_F is dominant and quasi-finite, its generic fiber is zero-dimensional, so $[L : K] < \infty$. The morphism remains étale at the generic point; hence the finite extension L/K is separable. $\square$

Proposition 3.2 (Finite planar Keller rigidity)**.** *If the planar Keller morphism $f_F\colon \mathbb{A}^2_{\mathbb{C}} \to \mathbb{A}^2_{\mathbb{C}}$ is finite, then F is a polynomial automorphism.*

Proof. Proposition 3.1 makes f_F étale. Thus a finite f_F is a connected finite étale cover of $\mathbb{A}^2_{\mathbb{C}}$. Finite-étale rigidity of the affine plane, Theorem 3.27, makes it an isomorphism. Its inverse is a morphism of affine schemes and hence is given by polynomials. □

Definition 3.3 (Planar generic degree)**.** The generic degree of F is

$$d_F := \deg_{\text{gen}}(F) := [L : K].$$

Because L/K is finite and separable, d_F is also the number of K-embeddings $L \hookrightarrow \overline{K}$, equivalently the number of geometric points in the generic fiber of f_F.

3.1 The planar secant projector

Let $\Delta = V(I_\Delta) \subseteq \operatorname{Spec} S$. For $a \in S$, write $a|_\Delta$ for its image under $S \to S/I_\Delta$, and apply this notation entrywise to matrices. Let

$$\mathbf{d} = \begin{pmatrix} x_1 - y_1 \\ x_2 - y_2 \end{pmatrix}, \qquad \mathbf{r} = \begin{pmatrix} P(x) - P(y) \\ Q(x) - Q(y) \end{pmatrix}.$$

For $H \in \mathbb{C}[x_1, x_2]$, let

$$D_1 H = \frac{H(x_1, x_2) - H(y_1, x_2)}{x_1 - y_1}, \qquad D_2 H = \frac{H(y_1, x_2) - H(y_1, y_2)}{x_2 - y_2}.$$

Both quotients belong to S. Let

$$M_F = \begin{pmatrix} D_1 P & D_2 P \\ D_1 Q & D_2 Q \end{pmatrix}.$$

Then

$$\mathbf{r} = M_F \mathbf{d}, \qquad M_F\big|_\Delta = JF.$$

Let $\delta_F = \det(M_F)$.

Proposition 3.4 (Planar secant annihilation)**.** *The secant determinant satisfies*

$$\delta_F I_\Delta \subseteq I_R, \qquad \delta_F\big|_\Delta = c.$$

Proof. Multiplying $\mathbf{r} = M_F \mathbf{d}$ by $\operatorname{adj}(M_F)$ gives

$$\delta_F \mathbf{d} = \operatorname{adj}(M_F)\mathbf{r}.$$

$$\mathbf{r} \in I_R^{\oplus 2}, \qquad \operatorname{adj}(M_F) \in M_2(S) \quad \Longrightarrow \quad \operatorname{adj}(M_F)\mathbf{r} \in I_R^{\oplus 2}.$$

Therefore

$$\delta_F \mathbf{d} \in I_R^{\oplus 2},$$

or equivalently,

$$\forall j \in \{1, 2\}, \qquad \delta_F(x_j - y_j) \in I_R.$$

Hence $\delta_F I_\Delta \subseteq I_R$. On the diagonal,

$$(D_j F_i)|_\Delta = \frac{\partial F_i}{\partial x_j},$$

so $M_F|_\Delta = JF$ and $\delta_F|_\Delta = \det JF = c$. □

Let $\bar{\delta}_F$ be the class of δ_F in C_F, and define

$$q_F := 1 - c^{-1}\bar{\delta}_F.$$

Set also

$$e_F := 1 - q_F = c^{-1}\bar{\delta}_F.$$

Proposition 3.5 (Secant separability idempotent)**.** *Under $C_F \cong A \otimes_B A$, the element e_F satisfies*

$$e_F^2 = e_F, \qquad \bar{\mu}_F(e_F) = 1, \qquad (a \otimes 1)e_F = (1 \otimes a)e_F \quad (a \in A).$$

Thus e_F is the diagonal separability idempotent written explicitly by the planar secant determinant.

Proof. Proposition 3.4 gives $\bar{\delta}_F J = 0$, while $\bar{\delta}_F - c \in J$. Hence $\bar{\delta}_F(\bar{\delta}_F - c) = 0$, so $e_F^2 = e_F$. Diagonal evaluation gives $\bar{\mu}_F(\bar{\delta}_F) = c$, so $\bar{\mu}_F(e_F) = 1$. Finally, $a \otimes 1 - 1 \otimes a$ belongs to $J = \ker(\bar{\mu}_F)$, while e_F annihilates J; this is exactly the last displayed identity. □

Theorem 3.6 (Planar collision-sheet decomposition)**.** *The collision scheme decomposes as*

$$R_F \cong \Delta_{X/Y}(X) \amalg R_F^\circ, \qquad C_F \cong A \times C_F^\circ.$$

The off-diagonal ideal satisfies

$$I_{\text{off}} = I_R + (\delta_F) = I_R : I_\Delta^\infty.$$

Proof. By Proposition 2.2,

$$J := \ker(\bar{\mu}_F) = I_\Delta / I_R = \text{Obs}(F).$$

Proposition 3.4 gives

$$\bar{\delta}_F J = 0, \qquad \bar{\delta}_F - c \in J.$$

Theorem 2.5, applied to these two relations and $q_F = 1 - c^{-1}\bar{\delta}_F$, yields

$$q_F^2 = q_F, \qquad J = C_F q_F, \qquad \text{Ann}_{C_F}(J) = C_F(1 - q_F), \qquad J = 0 \Longleftrightarrow q_F = 0.$$

Because $1 - q_F = c^{-1}\bar{\delta}_F$,

$$\frac{I_R : I_\Delta}{I_R} = \text{Ann}_{C_F}(J) = C_F(1 - q_F) = C_F\bar{\delta}_F = \frac{I_R + (\delta_F)}{I_R}.$$

Hence

$$I_{\text{off}} = I_R : I_\Delta = I_R + (\delta_F).$$

Idempotence gives $J^m = C_F q_F = J$ for every $m \geq 1$. Therefore

$$\frac{I_R : I_\Delta^m}{I_R} = \text{Ann}_{C_F}(J^m) = \text{Ann}_{C_F}(J) = \frac{I_R : I_\Delta}{I_R}.$$

Thus

$$I_R : I_\Delta^\infty = \bigcup_{m \geq 1} (I_R : I_\Delta^m) = I_R : I_\Delta.$$

The idempotent q_F also gives a ring isomorphism

$$C_F \xrightarrow{\sim} C_F(1-q_F) \times C_F q_F, \qquad a \longmapsto (a(1-q_F), aq_F),$$

whose inverse is addition. The two factors satisfy

$$C_F(1-q_F) \cong C_F/C_F q_F = C_F/J \xrightarrow[\bar{\mu}_F]{\sim} A$$

and

$$C_F q_F \cong C_F/C_F(1-q_F) \cong S/\big(I_R + (\delta_F)\big) = C_F^\circ.$$

Thus $C_F \cong A \times C_F^\circ$. Since the spectrum of a product is the disjoint union of the two spectra, this isomorphism dualizes to

$$R_F \cong \Delta_{X/Y}(X) \amalg R_F^\circ.$$

□

Corollary 2.4 states that

$$R_F^\circ = \varnothing \iff C_F^\circ = 0 \iff \mathrm{Obs}(F) = 0 \iff I_R = I_\Delta.$$

The preceding theorem identifies

$$C_F^\circ = S/\big(I_R + (\delta_F)\big),$$

and Theorem 2.5 gives $\mathrm{Obs}(F) = 0 \iff q_F = 0$. Hence

$$q_F = 0 \iff I_R + (\delta_F) = S \iff R_F^\circ = \varnothing \iff I_R = I_\Delta.$$

In particular, the secant determinant cuts out the off-diagonal factor:

$$R_F^\circ = \mathrm{Spec}\big(S/(I_R + (\delta_F))\big).$$

The entries of the full secant matrix give more than its determinant. Write

$$M_F = \begin{pmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{pmatrix}, \qquad d_i = x_i - y_i, \qquad r_1 = P(x) - P(y), \quad r_2 = Q(x) - Q(y),$$

and define

$$\Phi_F = \begin{pmatrix} m_{22} & -m_{21} \\ -m_{12} & m_{11} \\ -d_1 & -d_2 \end{pmatrix}.$$

Proposition 3.7 (Secant Hilbert–Burch resolution)**.** *Assume that $C_F^\circ \neq 0$. The signed maximal minors of Φ_F are r_1, r_2, δ_F, and there is an exact sequence*

$$0 \longrightarrow S^2 \xrightarrow{\Phi_F} S^3 \xrightarrow{(r_1, r_2, \delta_F)} S \longrightarrow C_F^\circ \longrightarrow 0.$$

Consequently

$$\omega_{C_F^\circ} \cong \mathrm{coker}(\Phi_F^{\mathsf{t}}) \cong C_F^\circ.$$

The last isomorphism is induced by

$$(u, v) \longmapsto -d_2 u + d_1 v.$$

Proof. Direct calculation gives

$$\det\begin{pmatrix}-m_{12} & m_{11}\\ -d_1 & -d_2\end{pmatrix}=r_1,\quad -\det\begin{pmatrix}m_{22} & -m_{21}\\ -d_1 & -d_2\end{pmatrix}=r_2,\quad \det\begin{pmatrix}m_{22} & -m_{21}\\ -m_{12} & m_{11}\end{pmatrix}=\delta_F.$$

Thus the maximal-minor ideal is $I_R+(\delta_F)=I_{\mathrm{off}}$. The first projection makes $C_F=A\otimes_B A$ an étale A-algebra, so its nonzero direct factor C_F° is a regular Cohen–Macaulay surface. Hence $I_{\mathrm{off}}\subset S$ has grade two, and Hilbert–Burch gives the displayed resolution [The26q].

Dualizing the resolution over the regular four-dimensional ring S gives $\omega_{C_F^\circ}\cong\operatorname{coker}(\Phi_F^{\mathsf{t}})$. The row $(-d_2,d_1)$ kills every row of Φ_F modulo (r_1,r_2,δ_F), so it induces the stated map to C_F°. Moreover $(d_1,d_2)C_F^\circ=C_F^\circ$: otherwise a prime containing both would also contain δ_F, although $\delta_F\equiv c \pmod{(d_1,d_2)}$. The induced map is therefore surjective. Both source and target are rank-one locally free modules on the regular scheme $\operatorname{Spec}C_F^\circ$, so it is an isomorphism. $\square$

The final trivialization is intrinsic up to a unit. Identifying this particular Hilbert–Burch generator with a chosen volume form requires a separate comparison; no such identification is used below.

3.2 Galois specialization and conjugate coordinates

Fix an algebraic closure $\overline{K}\supseteq L$, and let

$$\begin{aligned}
N &:= K\big(\sigma(L):\sigma\in\operatorname{Hom}_K(L,\overline{K})\big), & K&\subseteq L\subseteq N\subseteq\overline{K},\\
\overline{A} &:= \{a\in L: a \text{ is integral over } B\}, & A_N&:=\{a\in N: a\text{ is integral over } B\},\\
\overline{X} &:= \operatorname{Spec}\overline{A}, & Z&:=\operatorname{Spec}A_N.
\end{aligned}$$

Thus N/K is the normal closure of L/K.

Proposition 3.8 (Planar finite-cover diagram)**.**

$$Z\xrightarrow[\text{finite}]{\pi}\overline{X}\xrightarrow[\text{finite}]{\bar\nu}Y,\qquad \nu=\bar\nu\circ\pi,$$

and

$$X\xhookrightarrow[\text{open}]{j}\overline{X},\qquad \bar\nu\circ j=f_F.$$

Proof. Since $B\cong\mathbb{C}[u,v]$ is a finitely generated normal $\mathbb{C}$-algebra, its integral closures in the finite extensions L/K and N/K are finite over B. The inclusion $L\subseteq N$ gives $\overline{A}\subseteq A_N$, so

$$B\subseteq\overline{A}\subseteq A_N.$$

The morphism $\bar\nu$ is finite because $\overline{A}$ is a finite B-module. If $a_1,\dots,a_r$ generate A_N over B, then they also generate A_N over $\overline{A}$, since $B\subseteq\overline{A}$. Hence π is finite.

For $a\in\overline{A}$, integrality over $B\subseteq A$ and the normality of $A=\mathbb{C}[x_1,x_2]$ give $a\in A$. Thus

$$B\subseteq\overline{A}\subseteq A,$$

and $\operatorname{Frac}(\overline{A})=L$: clearing denominators in an algebraic equation over $K=\operatorname{Frac}(B)$ shows that for every $\ell\in L$, some nonzero $b\in B$ makes $b\ell$ integral over B. Thus $\ell=(b\ell)/b\in\operatorname{Frac}(\overline{A})$. The inclusion $\overline{A}\subseteq A$ therefore induces a dominant birational morphism $j\colon X\to\overline{X}$.

The morphism j is quasi-finite: because $\bar\nu\circ j=f_F$, each fiber of j is contained in a fiber of the quasi-finite morphism f_F. It is separated because it is a morphism between affine schemes. Since $\overline{X}$ is normal, Zariski's Main Theorem makes j an open immersion. $\square$

One consequence of this finiteness is useful bookkeeping, but should not be confused with a vanishing theorem. If $E = V(\mathfrak{p}_E) \subset Z$ is a prime divisor and $D = V(\mathfrak{p}_E \cap B) \subset Y$ is its image, then

$$A_N/\mathfrak{p}_E \quad \text{is finite over} \quad B/(\mathfrak{p}_E \cap B).$$

Thus $E \to D$ is a finite map of affine curves. Neither coordinate ring is thereby finite-dimensional over $\mathbb{C}$, and this curvewise finiteness does not bound poles in the direction transverse to E. More precisely, if $e(E/D)$ is the DVR ramification index and $f(E/D) = [\kappa(E) : \kappa(D)]$ is the degree of the induced curve map, then

$$\sum_{E|D} e(E/D)f(E/D) = [N : K].$$

Riemann–Hurwitz or equal-genus arguments for compactifications of $E \to D$ constrain $f(E/D)$; the inertia calculation in this paper requires control of the distinct transverse index $e(E/D)$. This is why curvewise finiteness alone does not eliminate a ramified divisor.

3.3 Global and divisorial formulations

Let $\mathfrak{b} \subseteq \overline{A}$ be the radical ideal defining $\partial X = \overline{X} \setminus j(X)$. Because $\overline{X}$ is affine, local cohomology with support in the boundary is

$$H^1_{\partial X}(\overline{X}, \mathcal{O}_{\overline{X}}) = H^1_{\mathfrak{b}}(\overline{A}).$$

Definition 3.9 (Global and divisorial finiteness formulations)**.** For a planar Keller map F, define:

1. The *global boundary formulation*, PlanarBoundaryCoherence, denoted $\mathrm{PBC}(F)$, is the assertion that

$$H^1_{\mathfrak{b}}(\overline{A}) \quad \text{is a finite (equivalently, finitely generated) } \overline{A}\text{-module.}$$

2. The *divisorial differential formulation*, PlanarRamificationRigidity, denoted $\mathrm{PRR}(F)$, is the assertion that

$$\Omega_{A_N/B} \quad \text{has finite length as an } A_N\text{-module.}$$

Here finiteness does not repeat the finiteness already proved in Proposition 3.8. The rings $\overline{A}$ and A_N are finite B-modules because they are integral closures in finite field extensions. Consequently $\overline{X}$ and Z are reduced, irreducible affine schemes of finite type over the algebraically closed field $\mathbb{C}$, hence affine varieties in the usual sense. Ordinary higher cohomology of quasi-coherent sheaves on these affine schemes vanishes (the affine-cohomology theorem, or the vanishing half of Serre's affineness criterion) [The26i]. Neither fact implies that the boundary local cohomology $H^1_{\mathfrak{b}}(\overline{A})$ is finite: local cohomology measures sections with poles along the omitted closed set. Similarly, $\Omega_{A_N/B}$ is automatically a finite A_N-module, because A_N is a finite-type B-algebra, but requiring it to have finite *length* is the stronger assertion that its support has dimension zero.

Proposition 3.10 (The boundary-coherence implication)**.** *Planar boundary coherence gives the implication chain*

$$\mathrm{PBC}(F) \Longrightarrow \partial X = \varnothing \Longrightarrow f_F \text{ is finite étale} \Longrightarrow F \text{ is a polynomial automorphism.}$$

Proof. Because $\overline{A}$ is a domain and $j(X)$ contains its generic point, $H^0_{\mathfrak{b}}(\overline{A}) = 0$. The local-cohomology sequence for $j(X) \subseteq \overline{X}$, together with affineness of $\overline{X}$, therefore gives

$$0 \longrightarrow \overline{A} \longrightarrow A \longrightarrow H^1_{\mathfrak{b}}(\overline{A}) \longrightarrow 0.$$

Here $\Gamma(X, \mathcal{O}_X) = A$; this is the standard affine open-complement local-cohomology sequence [The26l]. If the last term is a finite $\overline{A}$-module, then so is A. Thus the open immersion j is finite. Its image is both open and closed; since $\overline{X}$ is integral and $X \neq \varnothing$, it is all of $\overline{X}$. Hence $\partial X = \varnothing$, so $f_F = \bar{\nu}$ is finite. It is already étale by Proposition 3.1, and Proposition 3.2 finishes the proof. □

In particular, $\mathrm{PBC}(F)$ implies $\mathrm{PRR}(F)$: after F is an automorphism one has $A_N = B$, so $\Omega_{A_N/B} = 0$. The converse does not follow by a direct module-theoretic comparison. The ramification-rigidity endgame below nevertheless shows that, for a planar Keller map, $\mathrm{PRR}(F)$ also implies automorphy and hence $\mathrm{PBC}(F)$. The two criteria record different mechanisms: ramification rigidity first eliminates divisorial ramification, whereas boundary coherence eliminates the entire deleted boundary at once.

Let $S_F \subseteq Y$ denote the nonproperness set: the complement of the largest open subset $U \subseteq Y$ over which $f_F^{-1}(U) \to U$ is proper.

Proposition 3.11 (The deleted boundary and properness)**.** *Set-theoretically,*

$$S_F = \bar{\nu}(\partial X).$$

Consequently, the following are equivalent:

$$f_F \text{ is proper}, \qquad S_F = \varnothing, \qquad \partial X = \varnothing.$$

Under any of these conditions, F is a polynomial automorphism.

Proof. The set $\bar{\nu}(\partial X)$ is closed because $\bar{\nu}$ is finite. Over

$$U = Y \setminus \bar{\nu}(\partial X),$$

the open immersion identifies $f_F^{-1}(U)$ with $\bar{\nu}^{-1}(U)$; hence the restriction of f_F is finite and therefore proper. Thus $S_F \subseteq \bar{\nu}(\partial X)$.

Conversely, suppose that $y \in \bar{\nu}(\partial X)$ and that f_F were proper over a neighborhood of y. Since f_F is quasi-finite, it would be finite after shrinking to an affine neighborhood V of y. The normal finite V-scheme $f_F^{-1}(V)$, with function field L, would then be the normalization of V in L. By uniqueness of normalization it would equal $\bar{\nu}^{-1}(V)$, contrary to the existence of a boundary point above y. Therefore $\bar{\nu}(\partial X) \subseteq S_F$, proving the equality and the three equivalences.

If they hold, $f_F = \bar{\nu}$ is finite and étale. Since $Y \cong \mathbb{A}^2_{\mathbb{C}}$, Theorem 3.27 makes this connected cover trivial, so F is a polynomial automorphism. □

Thus the deleted normalization boundary is not auxiliary: it is precisely the geometric source of nonproperness. For a nonsingular polynomial map of the complex plane, a nonempty S_F is known to be a curve with one point at infinity [Cha04]. The inertia calculations below identify which divisorial part of this nonproperness can support the normal closure.

Here dimension two materially sharpens the normalization problem. Both $\overline{X}$ and Z are normal Noetherian surfaces finite over the regular surface Y. A normal Noetherian scheme of dimension at most two is Cohen–Macaulay [The26a]; since the finite maps have zero-dimensional fibers, miracle flatness makes them finite flat [The26m]. Their coordinate rings are therefore finite locally free B-modules, so the trace, different, and ramification divisor belong to the finite-flat surface setting.

Moreover, $j(X)$ is a dense affine open in the separated Noetherian surface $\overline{X}$. Every irreducible component of $\partial X = \overline{X} \setminus j(X)$ consequently has codimension one [The26b]: a nonempty deleted boundary is a union of curves. Purity of the branch locus rules out ramification supported only in codimension two. Thus the remaining planar question is the following concrete one: can a nonzero ramification divisor on Z have every inertia-moving conjugate center on the deleted boundary curves? The coordinate criterion below gives an exact valuation formulation of this question.

Characteristic zero also makes the ramification divisor of the intermediate normalization explicit. Write

$$R_{\bar{\nu}} = \sum_{C \in \overline{X}^{(1)}} (e(C/\bar{\nu}(C)) - 1)C.$$

The different exponent is $e - 1$ at each divisorial valuation, and the étaleness of f_F gives

$$\operatorname{Supp}(R_{\bar{\nu}}) \subseteq \partial X.$$

In canonical-divisor language, $R_{\bar{\nu}}$ is the relative canonical divisor of the finite map. Its vanishing is therefore the finite-cover analogue of crepancy; this is a ramification statement, not a consequence of resolving the isolated singularities of the normal surface $\overline{X}$.

Proposition 3.12 (Boundary divisor-class rigidity)**.** *Let D be an effective Weil divisor on $\overline{X}$ supported on ∂X. If $[D]$ is torsion in $\mathrm{Cl}(\overline{X})$, then $D = 0$.*

Proof. Choose $m > 0$ and $h \in L^\times$ with

$$\operatorname{div}_{\overline{X}}(h) = mD.$$

The divisor on the right is effective. Since $\overline{X}$ is normal, the height-one valuation criterion for regularity gives $h \in \Gamma(\overline{X}, \mathcal{O}_{\overline{X}}) = \overline{A}$. Because D is supported on the deleted boundary, $\operatorname{div}_X(h|_X) = 0$. Since $A = \mathbb{C}[x_1, x_2]$ is a unique factorization domain, this gives

$$h|_X \in A^\times = \mathbb{C}[x_1, x_2]^\times = \mathbb{C}^\times.$$

The inclusion $\overline{A} \subseteq A$ is injective, so h itself is constant. Therefore $mD = 0$, and hence $D = 0$. $\square$

Corollary 3.13 (Different-class criterion for planar rigidity)**.** *If $[R_{\bar{\nu}}]$ is torsion in $\mathrm{Cl}(\overline{X})$, then F is a polynomial automorphism. In particular, this holds if the different is principal or if $\mathrm{Cl}(\overline{X})$ is torsion.*

Proof. Proposition 3.12 gives $R_{\bar{\nu}} = 0$. Purity of the branch locus then makes the finite map $\bar{\nu}$ étale, and finite-étale rigidity of $\mathbb{A}^2_{\mathbb{C}}$ makes it an isomorphism. Thus $L = K$. Proposition 3.1 gives that A is flat over B, so Corollary 2.17 gives $\operatorname{Obs}(F) = 0$. Theorem 2.15 finishes the proof. $\square$

The following specialization recovers the classical Galois case of the Jacobian conjecture [Cam73, Raz79, Wri81] in the boundary language of this section.

Theorem 3.14 (Boundary-theoretic Galois rigidity)**.** *If the marked function-field extension L/K is normal, equivalently Galois, then*

$$L = K \qquad \text{and} \qquad F \in \mathrm{Aut}_{\mathrm{poly}}(\mathbb{A}^2_{\mathbb{C}}).$$

Proof. Proposition 3.1 shows that L/K is finite and separable, so normality is equivalent to the Galois condition. Let $B_1, \ldots, B_r$ be the prime divisors of Y that are images of components of $R_{\bar{\nu}}$. Since $B = \mathbb{C}[P, Q]$ is a unique factorization domain, choose an irreducible $b_i \in B$ such that

$$B_i = \mathrm{div}_Y(b_i).$$

The Galois group preserves the integral closure $\overline{X}$ and acts transitively on the prime divisors above a fixed B_i, so their ramification indices have a common value e_i [The26h]. Put

$$D_i = \sum_{\substack{C \in \overline{X}^{(1)} \\ \bar{\nu}(C) = B_i}} C.$$

For every $C \mid B_i$, the valuation formula for pullback gives

$$\mathrm{ord}_C(b_i) = e(C/B_i)\,\mathrm{ord}_{B_i}(b_i) = e_i.$$

There are no other height-one zeros of b_i on $\overline{X}$, and hence

$$\mathrm{div}_{\overline{X}}(b_i) = \bar{\nu}^* B_i = e_i D_i.$$

These are the standard Galois valuation and divisor-pullback formulas [The26f, The26o].

The residue characteristic is zero, so the divisorial extensions are tame and the different exponent at every $C \mid B_i$ is $e_i - 1$ [The26p]. Grouping the components of the ramification divisor by their images therefore gives

$$R_{\bar{\nu}} = \sum_{i=1}^{r} (e_i - 1) D_i.$$

Set $m = \prod_{i=1}^{r} e_i$, with the empty product understood as 1, and define

$$h = \prod_{i=1}^{r} b_i^{m(e_i-1)/e_i} \in K^\times \subseteq L^\times.$$

Every exponent is a nonnegative integer, and the preceding pullback formula yields

$$\mathrm{div}_{\overline{X}}(h) = \sum_{i=1}^{r} \frac{m(e_i - 1)}{e_i}\, e_i D_i = m R_{\bar{\nu}}.$$

Thus $[R_{\bar{\nu}}]$ is torsion in $\mathrm{Cl}(\overline{X})$, and Corollary 3.13 gives the result. $\square$

Corollary 3.15 (Quadratic planar rigidity)**.** *No planar Keller map has generic degree $d_F = 2$. Consequently,*

$$d_F \le 2 \quad \Longrightarrow \quad d_F = 1 \quad \Longrightarrow \quad F \in \mathrm{Aut}_{\mathrm{poly}}(\mathbb{A}^2_{\mathbb{C}}).$$

In particular, every nonautomorphic planar Keller map would have

$$d_F \ge 3 \qquad \textit{and} \qquad L/K \ \textit{nonnormal}.$$

Proof. If $d_F = 2$, then the separable quadratic extension L/K is Galois. Theorem 3.14 would give $L = K$, contradicting $[L : K] = 2$. Hence $d_F \le 2$ forces $d_F = 1$; Corollary 2.17 and Theorem 2.15 give automorphy. Finally, Theorem 3.14 rules out normality in every nonautomorphic case. $\square$

Outside this normal/Galois specialization, the divisor-theoretic problem is to determine whether the finite-flat trace and different, together with $\overline{A} \subseteq \mathbb{C}[x_1, x_2]$, force $[R_{\bar{\nu}}]$ to be torsion. The different-class criterion proves the consequence of such torsion; it does not assert it for a general nonnormal extension. Corollary 3.15 shows that only nonnormal extensions of generic degree at least three remain.

It is important that this argument cannot simply be rerun for the always Galois extension N/K. Boundary divisor-class rigidity applies on $\overline{X}$ because it contains the marked affine plane $X \simeq \mathbb{A}^2_{\mathbb{C}}$ as a dense open. If L/K is nonnormal, the Galois normalization Z instead maps finitely to every conjugate intermediate model and need not contain a marked affine-plane open whose deleted boundary supports all of $R_{Z/Y}$. This is precisely why the pulled-back conjugate boundaries and moving-sheet condition are needed.

Let

$$\mathcal{D}_F = \left(Z \xrightarrow{\pi} \overline{X} \xrightarrow{\bar{\nu}} Y,\, j\colon X \hookrightarrow \overline{X}\right), \qquad \partial X = \overline{X} \setminus j(X),$$
$$G = \mathrm{Gal}(N/K), \qquad H = \mathrm{Gal}(N/L).$$

The abstract conjugate centers now admit a direct two-coordinate test. For a representative $g \in G$ of $\omega = D_E gH$, the conjugate affine sheet is generated inside N by

$$A_g = \mathbb{C}[g(x_1), g(x_2)].$$

Proposition 3.16 (Planar coordinate test for a conjugate center)**.** *For $E \in Z^{(1)}$ and $\omega = D_E gH \in \mathcal{Q}_E$,*

$$\mathrm{cent}(E, \omega) \in j(X) \iff w_E(g(x_1)) \ge 0 \text{ and } w_E(g(x_2)) \ge 0.$$

Consequently,

$$\mathrm{cent}(E, \omega) \in \partial X \iff \min\{w_E(g(x_1)), w_E(g(x_2))\} < 0.$$

Proof. By definition, $u_\omega = (g^{-1} w_E)|_L$, so

$$u_\omega(x_i) = w_E(g(x_i)) \qquad (i = 1, 2).$$

The center of u_ω lies on the affine open $X = \operatorname{Spec} \mathbb{C}[x_1, x_2]$ exactly when its valuation ring contains A. Since A is generated by x_1, x_2, this is equivalent to the two displayed nonnegativity conditions. Negating them gives the boundary criterion. $\square$

The early secant estimate has a precise interpretation on these conjugate generic sheets. For $g \in G$, let

$$\delta_{F,g} := \delta_F(x_1, x_2, g(x_1), g(x_2)) \in N, \qquad q_{F,g} := 1 - c^{-1}\delta_{F,g}.$$

Proposition 3.17 (Secant evaluation on conjugate sheets)**.** *For every $g \in G$,*

$$\delta_{F,g}(x_i - g(x_i)) = 0 \qquad (i = 1, 2).$$

Consequently,

$$\delta_{F,g} = \begin{cases} c, & g \in H, \\ 0, & g \notin H, \end{cases} \qquad q_{F,g} = \begin{cases} 0, & g \in H, \\ 1, & g \notin H. \end{cases}$$

Thus the planar secant projector is exactly the characteristic function of the nonmarked generic conjugate sheets.

Proof. The marked embedding $L \hookrightarrow N$ and its g-conjugate agree on K, so together they define an evaluation map from the generic collision algebra $L \otimes_K L$ to N. Evaluating Proposition 3.4 under this map gives the first display. If $g \in H = \mathrm{Gal}(N/L)$, then $g(x_i) = x_i$ for both coordinates, and diagonal evaluation gives $\delta_{F,g} = c$. If $g \notin H$, then some x_i is moved, since $L = \mathbb{C}(x_1, x_2)$. The corresponding factor $x_i - g(x_i)$ is nonzero in the field N, so the first display forces $\delta_{F,g} = 0$. The formulas for $q_{F,g}$ follow. □

The complementary idempotent $e_F = 1 - q_F$ gives an explicit trace-theoretic bridge. Under the standard finite Galois decomposition

$$N \otimes_K N \xrightarrow{\sim} \prod_{\sigma \in G} N, \qquad a \otimes b \longmapsto \big(a\,\sigma(b)\big)_\sigma,$$

the generic base change of e_F is the characteristic function of H. Let $e_g = (g \otimes g)e_F$; then e_g is the characteristic function of gHg^{-1}. Proposition 2.8 gives

$$e_N := \prod_{g \in G} e_g = \mathbf{1}_{\mathrm{core}_G(H)} = \mathbf{1}_{\{1\}}.$$

Finally, the trace pairing gives a canonical K-linear isomorphism

$$\Theta_N : N \otimes_K N \xrightarrow{\sim} \mathrm{End}_K(N), \qquad \Theta_N(a \otimes b)(z) = a\,\mathrm{Tr}_{N/K}(bz),$$

and $\Theta_N(e_N) = \mathrm{id}_N$. Thus the product of the conjugate planar secant idempotents is the trace coevaluation tensor for N/K. Notice that Θ_N is a linear, not a ring, isomorphism. For the finite locally free order $T = A_N$, the integral coevaluation tensor lies in $T \otimes_B \mathrm{Hom}_B(T, B)$; its generic fiber is e_N after the trace identification. This identifies the finite trace-dual lattice canonically, but does not yet bound poles of moving-sheet principal parts.

This follows the exponent-one secant bound completely into the normal closure, but also locates its limit. On a nonmarked sheet $\delta_{F,g}$ is the zero function; it is therefore a generic-sheet projector, not a nonzero boundary parameter whose fixed power bounds pole order. Any use of the secant matrix for uniform boundary control therefore needs a further transport statement, using more than the determinant alone. That possible mechanism is isolated after the completed endgame.

For $E \in \mathrm{Ram}^{(1)}(Z/Y)$ and $\omega \in \mathcal{Q}_E$, Proposition 2.11 gives

$$e\big(\mathrm{cent}(E, \omega)/\nu(E)\big) = i_E(\omega).$$

Since $X \to Y$ is étale by Proposition 3.1, Proposition 2.12 gives

$$i_E(\omega) > 1 \quad \Longrightarrow \quad \mathrm{cent}(E, \omega) \in \partial X.$$

Thus Definition 2.13 specializes to

$$\mathcal{H}(\mathcal{D}_F) = \left\{ E \in \mathrm{Ram}^{(1)}(Z/Y) : \begin{array}{l} \exists \omega \in \mathcal{Q}_E,\ i_E(\omega) > 1, \\ \forall \omega \in \mathcal{Q}_E, \quad i_E(\omega) > 1 \Longrightarrow \mathrm{cent}(E, \omega) \in \partial X \end{array} \right\}.$$

Proposition 3.16 makes this locus fully explicit:

$$\mathcal{H}(\mathcal{D}_F) = \left\{ E \in \mathrm{Ram}^{(1)}(Z/Y) : \begin{array}{l} \forall g \in G, \\ [I_E : I_E \cap gHg^{-1}] > 1 \Longrightarrow \min\{w_E(g(x_1)), w_E(g(x_2))\} < 0 \end{array} \right\}.$$

The existence of at least one g with nontrivial relative index is automatic from $I_E \neq 1$ and $\mathrm{core}_G(H) = 1$. Thus hidden inertia is not an additional abstract object in the planar proof: it is precisely the possibility that every inertia-moving conjugate coordinate pair acquires a pole at the deleted boundary.

Definition 3.18 (Planar conjugate coverage)**.** Write $\mathrm{Cov}_2(F)$ for the following condition: every $E \in \mathrm{Ram}^{(1)}(Z/Y)$ admits a $g \in G$ such that

$$[I_E : I_E \cap gHg^{-1}] > 1, \qquad w_E(g(x_1)) \geq 0, \qquad w_E(g(x_2)) \geq 0.$$

Thus some conjugate sheet moved by inertia still has an affine center.

We now construct the canonical boundaries directly on the common Galois normalization. For $g \in G$, set

$$\overline{A}_g = g(\overline{A}), \qquad \overline{X}_g = \mathrm{Spec}\,\overline{A}_g, \qquad X_g = \mathrm{Spec}\,A_g.$$

The conjugate of j is an open immersion $X_g \hookrightarrow \overline{X}_g$, and the inclusion $\overline{A}_g \subseteq A_N$ gives a finite map $\pi_g \colon Z \to \overline{X}_g$. Let

$$U_g := \pi_g^{-1}(X_g) \subseteq Z, \qquad D_g := Z \setminus U_g, \qquad \mathfrak{j}_g := I_{A_N}(D_g) = \bigcap_{\mathfrak{p} \in D_g} \mathfrak{p}.$$

Thus $\mathfrak{j}_g$ is canonically the radical vanishing ideal of the pulled-back deleted boundary and $D_g = V(\mathfrak{j}_g)$. The construction depends only on gH, since right multiplication by H does not change the conjugate embedding of L.

Proposition 3.19 (Pulled-back boundary dictionary)**.** *For every height-one point E of Z and every $g \in G$,*

$$\eta_E \in U_g \quad \Longleftrightarrow \quad \mathfrak{j}_g \not\subseteq \mathfrak{p}_E \quad \Longleftrightarrow \quad w_E(g(x_1)) \geq 0 \quad \textit{and} \quad w_E(g(x_2)) \geq 0.$$

Equivalently,

$$\mathfrak{j}_g \subseteq \mathfrak{p}_E \quad \Longleftrightarrow \quad \eta_E \in D_g \quad \Longleftrightarrow \quad \min\{w_E(g(x_1)), w_E(g(x_2))\} < 0.$$

Proof. Because $D_g = V(\mathfrak{j}_g)$, membership $\eta_E \in D_g$ is exactly the containment $\mathfrak{j}_g \subseteq \mathfrak{p}_E$. The point η_E belongs to U_g precisely when its image under π_g is centered on $X_g = \mathrm{Spec}\,\mathbb{C}[g(x_1), g(x_2)]$. The valuation criterion for a center on an affine model says exactly that both conjugate coordinates have nonnegative valuation. Taking complements gives the second display. □

Proposition 3.20 (Moving-sheet coverage dictionary)**.** *For $E \in \mathrm{Ram}^{(1)}(Z/Y)$, put*

$$\mathfrak{j}_E^{\mathrm{mov}} := \sum_{\substack{g \in G \\ I_E \not\subseteq gHg^{-1}}} \mathfrak{j}_g.$$

Then

$$\exists g \in G, \quad I_E \not\subseteq gHg^{-1}, \quad \eta_E \in U_g \quad \Longleftrightarrow \quad \mathfrak{j}_E^{\mathrm{mov}} \not\subseteq \mathfrak{p}_E.$$

Thus eliminating E is exactly the failure of its moving-boundary ideal to be contained in its height-one prime.

Proof. A prime contains a sum of ideals exactly when it contains every summand. Apply Proposition 3.19 to each inertia-moving g. $\square$

For a nontrivial subgroup $I \leq G$, define its moving-boundary ideal by

$$\mathfrak{j}_I^{\mathrm{mov}} := \sum_{\substack{g \in G \\ I \nsubseteq gHg^{-1}}} \mathfrak{j}_g.$$

The sum is taken in the common Galois-normalization ring A_N; this is why the Galois closure enters only at this final specialization.

The same common ring gives a finite-group formulation which does not first choose a ramified divisor. For a subgroup $C \leq G$, let

$$\mathfrak{a}_C := \big(\sigma(t) - t : \sigma \in C,\ t \in A_N\big) \subseteq A_N.$$

This is the scheme-theoretic fixed-locus ideal for the action of C on Z. If $\mathfrak{p}_E \in \operatorname{Spec} A_N$, then

$$\mathfrak{a}_C \subseteq \mathfrak{p}_E \quad \Longleftrightarrow \quad C \leq I_E.$$

Indeed, the containment says that every element of C stabilizes $\mathfrak{p}_E$ and acts trivially on its residue field, which is the definition of inertia at E.

Definition 3.21 (Planar boundary separation)**.** Write $\mathrm{Sep}_2(F)$ for the following condition: for every prime-order subgroup $C \leq G$,

$$\operatorname{Spec}^{(1)}(A_N) \cap V(\mathfrak{a}_C + \mathfrak{j}_C^{\mathrm{mov}}) = \varnothing.$$

Equivalently,

$$\forall \mathfrak{p} \in \operatorname{Spec}^{(1)}(A_N), \qquad \mathfrak{a}_C \subseteq \mathfrak{p} \quad \Longrightarrow \quad \mathfrak{j}_C^{\mathrm{mov}} \nsubseteq \mathfrak{p}.$$

In codimension language, every fixed–moving boundary locus $V(\mathfrak{a}_C + \mathfrak{j}_C^{\mathrm{mov}})$ has codimension at least two in the normal surface Z. No assertion is made about isolated fixed boundary points.

Proposition 3.22 (Boundary separation is the divisorial criterion)**.** *For a planar Keller map,*

$$\mathrm{Sep}_2(F) \quad \Longleftrightarrow \quad \mathrm{Ram}^{(1)}(Z/Y) = \varnothing.$$

Proof. Assume boundary separation and suppose that $E \in \mathrm{Ram}^{(1)}(Z/Y)$. Choose a subgroup $C \leq I_E$ of prime order; such a subgroup exists by Cauchy's theorem for the nontrivial finite group I_E. Then $\mathfrak{a}_C \subseteq \mathfrak{p}_E$. If $C \nsubseteq gHg^{-1}$, then $I_E \nsubseteq gHg^{-1}$, so the relative inertia index on the sheet represented by gH is nontrivial. Proposition 2.12 places its center in the deleted boundary, and hence $\mathfrak{j}_g \subseteq \mathfrak{p}_E$. This holds for every sheet moved by C, so

$$\mathfrak{a}_C + \mathfrak{j}_C^{\mathrm{mov}} \subseteq \mathfrak{p}_E,$$

contrary to boundary separation.

Conversely, if boundary separation fails, there are a prime-order $C \leq G$ and a height-one prime $\mathfrak{p}_E$ containing $\mathfrak{a}_C + \mathfrak{j}_C^{\mathrm{mov}}$. The fixed-locus containment gives $C \leq I_E$. Since $C \neq 1$, the point E is ramified, contradicting $\mathrm{Ram}^{(1)}(Z/Y) = \varnothing$. $\square$

Proposition 3.23 (Coverage eliminates a ramified divisor)**.** *Let* $E \in \mathrm{Ram}^{(1)}(Z/Y)$. *If there is a* $g \in G$ *such that*

$$I_E \nsubseteq gHg^{-1} \qquad \textit{and} \qquad \eta_E \in U_g,$$

then a contradiction follows. Consequently,

$$\mathrm{Cov}_2(F) \quad \Longrightarrow \quad \mathrm{Ram}^{(1)}(Z/Y) = \varnothing.$$

Proof. The subgroup condition gives

$$i_E(gH) = [I_E : I_E \cap gHg^{-1}] > 1.$$

On the other hand, $\eta_E \in U_g$ places the conjugate center on the affine sheet X_g. The map $X_g \to Y$ is the Galois conjugate of the planar Keller map and is therefore étale. Its local ramification index at that center is one. Hence $1 < i_E(gH) = 1$, a contradiction. □

This is the point at which a height-one divisor is eliminated. Purity is not used in Proposition 3.23; it enters only after the proposition has been applied to every ramified height-one point.

3.4 Moving-sheet criteria for planar ramification rigidity

Corollary 3.24 (Planar hidden-inertia identification)**.**

$$\mathcal{H}(\mathcal{D}_F) = \mathrm{Ram}^{(1)}(Z/Y).$$

In particular,

$$\mathcal{H}(\mathcal{D}_F) = \varnothing \quad \Longleftrightarrow \quad \mathrm{Ram}^{(1)}(Z/Y) = \varnothing.$$

These conditions are also equivalent to planar conjugate coverage.

Proof. The equality is Proposition 2.14, applied to the planar finite-cover diagram. Conjugate coverage implies an empty ramification locus by Proposition 3.23, and hence an empty hidden-inertia locus by the displayed equality. Conversely, if the ramification locus is empty, conjugate coverage holds vacuously. □

Proposition 3.25 (The ramification-rigidity dictionary)**.** *For a planar Keller map,*

$$\mathrm{PRR}(F) \quad \Longleftrightarrow \quad \mathrm{Ram}^{(1)}(Z/Y) = \varnothing \quad \Longleftrightarrow \quad \mathrm{Cov}_2(F) \quad \Longleftrightarrow \quad \mathrm{Sep}_2(F).$$

Thus moving-sheet coverage and boundary separation are two exact criteria for PlanarRamificationRigidity, not additional global hypotheses.

Proof. The A_N-module $\Omega_{A_N/B}$ is finite, and its generic localization vanishes because N/K is separable. Since A_N is a two-dimensional affine $\mathbb{C}$-domain, every positive-dimensional closed subset has a height-one generic point. Thus this finite module has finite length exactly when it has no height-one point in its support. At a height-one point E, localizing B first at the image of E and then at $\mathfrak{p}_E$ gives the relevant finite local extension; the source local ring is a discrete valuation ring. In characteristic zero its residue extension is separable, so

$$\Omega_{(A_N)_{\mathfrak{p}_E}/B_{\mathfrak{p}_E \cap B}} = 0 \quad \Longleftrightarrow \quad I_E = 1.$$

Indeed, vanishing of the localized differential module is the finite-type unramifiedness criterion, and for a finite extension of discrete valuation rings in characteristic zero unramifiedness is equivalent to trivial inertia [The26j, The26f, The26o]. Consequently finite length is equivalent to the absence of divisorial ramification. Proposition 3.23 and vacuity in the unramified case give the equivalence with coverage; the equivalence with separation is Proposition 3.22. □

Equivalently, the premise $\mathcal{H}(\mathcal{D}_F) = \varnothing$ says that if a ramified divisor existed, some inertia-moving conjugate pair $g(x_1), g(x_2)$ would have to remain integral and hence centered on an affine sheet. Planar étaleness says that the same pair must lie at the boundary. This is the boundary contradiction used below; the premise supplies its exclusion, while the preceding results identify exactly what is being excluded.

3.5 The divisorial endgame

Theorem 3.26 (Purity of the branch locus, specialized)**.** *Let $Z \to \mathbb{A}^2_{\mathbb{C}}$ be a finite dominant generically separable morphism with Z integral and normal. If $\mathrm{Ram}^{(1)}(Z/\mathbb{A}^2_{\mathbb{C}}) = \varnothing$, then the morphism is étale.*

Proof. At every height-one point of Z, the absence of divisorial ramification gives ramification index one. The corresponding residue field extension is finite and, in characteristic zero, separable. Hence the morphism is unramified at every height-one point. Purity of the branch locus now implies that it is étale everywhere [The26n]. □

Theorem 3.27 (Finite-étale rigidity of the affine plane)**.** *Every connected finite étale cover of $\mathbb{A}^2_{\mathbb{C}}$ is trivial [GR71, Exposé XII, Théorème 5.1].*

Corollary 3.28 (Triviality of the planar normal closure)**.** *If*

$$\mathrm{Ram}^{(1)}(Z/Y) = \varnothing,$$

then

$$N = K \qquad \text{and hence} \qquad L = K.$$

Numerically,

$$[L : K] = [N : L] = [N : K] = 1.$$

Proof. By Proposition 3.1, $Y \cong \mathbb{A}^2_{\mathbb{C}}$. Since N/K is finite Galois, $\nu \colon Z \to Y$ is generically separable; hence Theorem 3.26 makes it finite étale. The inclusion

$$A_N \subset N \text{ is a domain,} \qquad \text{so} \qquad Z = \operatorname{Spec} A_N \text{ is connected.}$$

Theorem 3.27 therefore makes $\nu \colon Z \to Y$ an isomorphism. Taking function fields gives $N = K$. Finally, $K \subseteq L \subseteq N$ implies $L = K$. □

Corollary 3.29 (The divisorial endgame)**.** *The absence of divisorial ramification gives the implication chain*

$$\mathrm{Ram}^{(1)}(Z/Y) = \varnothing \Longrightarrow Z \longrightarrow Y \text{ is finite étale} \Longrightarrow N = K \Longrightarrow L = K \Longrightarrow q_F = 0 \Longrightarrow I_R = I_\Delta.$$

Proof. Purity, Theorem 3.26, makes $Z \to Y$ finite étale, and Corollary 3.28 gives $N = K$ and $L = K$. Thus $d_F = 1$. Flatness of A over B and Corollary 2.17 give $\mathrm{Obs}(F) = 0$. The abstract off-diagonal projector and collision criteria then give $q_F = 0$ and $I_R = I_\Delta$. □

By Proposition 3.25, any of $\mathrm{PRR}(F)$, $\mathrm{Sep}_2(F)$, or $\mathrm{Cov}_2(F)$ supplies the first condition in this single endgame. They are respectively differential, ideal-theoretic, and valuation-theoretic descriptions of its codimension-one premise, rather than three additional routes.

3.6 Vanishing of the off-diagonal factor

The main argument can now be read as a single spine:

$$\mathrm{Sep}_2(F) \Longleftrightarrow \mathrm{Cov}_2(F) \Longleftrightarrow \mathrm{Ram}^{(1)}(Z/Y) = \varnothing,$$
$$\mathrm{Ram}^{(1)}(Z/Y) = \varnothing \Longrightarrow Z/Y \text{ finite étale} \Longrightarrow N = L = K,$$
$$N = L = K \Longrightarrow F \in \mathrm{Aut}_{\mathrm{poly}}(\mathbb{A}^2_{\mathbb{C}}) \Longleftrightarrow I_R = I_\Delta.$$

The next theorem records the remaining named objects as equivalent translations of this spine.

Theorem 3.30 (Equivalent boundary criteria for planar vanishing)**.** *For every polynomial map $F = (P, Q)\colon \mathbb{A}^2_{\mathbb{C}} \to \mathbb{A}^2_{\mathbb{C}}$ satisfying $\det JF = c \in \mathbb{C}^\times$, the following conditions are equivalent:*

$$\mathrm{PBC}(F) \quad\Longleftrightarrow\quad \mathrm{PRR}(F) \quad\Longleftrightarrow\quad \mathrm{Sep}_2(F) \quad\Longleftrightarrow\quad \mathrm{Cov}_2(F)$$
$$\Longleftrightarrow\quad \mathcal{H}(\mathcal{D}_F) = \varnothing \quad\Longleftrightarrow\quad \mathrm{Ram}^{(1)}(Z/Y) = \varnothing,$$

and

$$\mathrm{Ram}^{(1)}(Z/Y) = \varnothing \quad\Longleftrightarrow\quad d_F = 1 \quad\Longleftrightarrow\quad \mathrm{Obs}(F) = 0,$$
$$\mathrm{Obs}(F) = 0 \quad\Longleftrightarrow\quad q_F = 0 \quad\Longleftrightarrow\quad R_F^\circ = \varnothing \quad\Longleftrightarrow\quad I_R = I_\Delta.$$

Proof. Proposition 3.10 shows that $\mathrm{PBC}(F)$ implies that F is an automorphism. Then $B = A$, $K = L = N$, $\partial X = \varnothing$, and $\Omega_{A_N/B} = 0$, so $\mathrm{PRR}(F)$ holds. Conversely, the argument below shows that $\mathrm{PRR}(F)$ implies automorphy; for an automorphism $\partial X = \varnothing$, and hence $H^1_{\partial X}(\overline{X}, \mathcal{O}_{\overline{X}}) = 0$. Thus $\mathrm{PBC}(F)$ holds as well.

Proposition 3.25 identifies $\mathrm{PRR}(F)$ and $\mathrm{Sep}_2(F)$ with the absence of divisorial ramification. Proposition 3.23 shows directly that $\mathrm{Cov}_2(F)$ implies $\mathrm{Ram}^{(1)}(Z/Y) = \varnothing$; conversely, if the ramification locus is empty, coverage holds vacuously. Corollary 3.24 identifies these conditions with $\mathcal{H}(\mathcal{D}_F) = \varnothing$.

Under these conditions, Corollary 3.28 gives $N = L = K$, hence $d_F = 1$, while Proposition 3.1 makes A flat over B. Corollary 2.17 therefore gives $\mathrm{Obs}(F) = 0$. Conversely, if $d_F = 1$, then $L = K$; flatness and Corollary 2.17 again give $\mathrm{Obs}(F) = 0$. If $\mathrm{Obs}(F) = 0$, then $I_R = I_\Delta$, so Theorem 2.15 makes F a polynomial automorphism. Hence $B = A$, $K = L = N$, and $\mathrm{Ram}^{(1)}(Z/Y) = \varnothing$.

Corollary 2.4 and Theorem 2.5 state

$$R_F^\circ = \varnothing \quad\Longleftrightarrow\quad C_F^\circ = 0 \quad\Longleftrightarrow\quad \mathrm{Obs}(F) = 0 \quad\Longleftrightarrow\quad I_R = I_\Delta,$$
$$\mathrm{Obs}(F) = 0 \quad\Longleftrightarrow\quad q_F = 0.$$

This proves the remaining equivalences. □

Corollary 3.31 (Boundary witness for failure of coverage)**.** *The equivalent conditions of Theorem 3.30 fail if and only if there is an $E \in \mathrm{Ram}^{(1)}(Z/Y)$ such that*

$$\forall g \in G, \qquad [I_E : I_E \cap gHg^{-1}] > 1 \quad\Longrightarrow\quad \min\{w_E(g(x_1)), w_E(g(x_2))\} < 0.$$

For such an E, at least one inertia-moving g exists. Thus failure of planar collision vanishing is witnessed by a ramified divisor whose inertia-moving conjugate centers all lie on the deleted boundary. By Theorem 3.14 and Corollary 3.15, its marked extension L/K is nonnormal and its generic degree satisfies $d_F \geq 3$.

Proof. If the equivalent conditions fail, then conjugate coverage fails. Negating Definition 3.18 supplies an $E \in \mathrm{Ram}^{(1)}(Z/Y)$ for which every g of nontrivial relative index fails at least one valuation inequality, giving the displayed condition. Conversely, the displayed condition is incompatible with a coverage witness for E, so conjugate coverage fails. Core-freeness guarantees the existence of at least one g of nontrivial relative index. □

Corollary 3.32 (Exact boundary forms of the planar Jacobian conjecture)**.** *The following universal statements are equivalent:*

1. *The classical complex two-dimensional Jacobian conjecture holds: every polynomial self-map of $\mathbb{A}^2_{\mathbb{C}}$ with constant nonzero Jacobian determinant is a polynomial automorphism.*
2. *Every planar Keller map satisfies* $\mathrm{PBC}(F)$*.*
3. *Every planar Keller map satisfies* $\mathrm{PRR}(F)$*.*
4. *Every planar Keller map satisfies* $\mathrm{Sep}_2(F)$*.*
5. *Every planar Keller map satisfies* $\mathrm{Cov}_2(F)$*.*

Equivalently, condition (4) *says that for every planar Keller map and every prime-order subgroup* $C \leq \mathrm{Gal}(N/K)$*,*

$$\operatorname{codim}_Z V(\mathfrak{a}_C + \mathfrak{j}_C^{\mathrm{mov}}) \geq 2.$$

Proof. If $JC(2)$ holds, every planar Keller map is an automorphism. Then $B = A$, $K = L = N$, $\overline{X} = X$, and $\partial X = \varnothing$, so both $H^1_{\partial X}(\overline{X}, \mathcal{O}_{\overline{X}})$ and $\Omega_{A_N/B}$ vanish. Thus (1) implies (2) and (3). Conversely, Proposition 3.10 shows that (2) implies (1). Theorem 3.30 identifies, for every planar Keller map,

$$\mathrm{PRR}(F) \iff \mathrm{Sep}_2(F) \iff \mathrm{Cov}_2(F) \iff \mathrm{Obs}(F) = 0.$$

Corollary 2.16 identifies universal obstruction vanishing with classical $JC(2)$, proving the remaining equivalences. The prime-order ideal formulation is Proposition 3.22. □

The planar section therefore gives an exact codimension-one reformulation. Here "divisorially coprime" means that $\mathfrak{a}_C$ and $\mathfrak{j}_C^{\mathrm{mov}}$ are contained in no common height-one prime; it does not mean that their sum is the unit ideal. Because Z is an integral surface, the relevant locus has only two possibilities: it has a codimension-one component, which is precisely the bad ramified-divisor case, or it has codimension at least two, in which case only isolated closed points may remain (or the locus is empty). There is no nonempty codimension-three case on Z, and purity is why these possible codimension-two points need not be eliminated separately.

In the good case, conjugate coverage excludes complete boundary-supported inertia orbits and the collision relation collapses to the diagonal. In the bad case, Corollary 3.31 produces exactly such a divisorial boundary witness. PlanarBoundaryCoherence records the equivalent global-boundary formulation that removes the entire normalization boundary at once.

3.7 Research mechanisms and status

The preceding equivalence theorem is the completed logical dictionary. The following mechanisms ask how its divisorial premise might be established from the special polynomial-plane data; they are not additional steps in the endgame.

For a normal Noetherian integral surface $W = \operatorname{Spec} R$ with dense open $U = W \setminus V(\mathfrak{j})$, define its boundary principal-parts module by

$$\mathcal{P}_{\mathfrak{j}}(R) := H^1_{\mathfrak{j}}(R).$$

The open-complement sequence identifies it with rational functions regular on U modulo those extending to W. At a height-one prime,

$$\mathcal{P}_{\mathfrak{j}}(R)_{\mathfrak{p}} \cong \begin{cases} 0, & \mathfrak{j} \not\subseteq \mathfrak{p}, \\ \operatorname{Frac}(R_{\mathfrak{p}})/R_{\mathfrak{p}}, & \mathfrak{j} \subseteq \mathfrak{p}. \end{cases}$$

Indeed, localization reduces to the one-element Čech complex of the DVR $R_{\mathfrak{p}}$. For a nontrivial $I \leq G$, put

$$\begin{aligned} U_I^{\mathrm{mov}} &:= Z \setminus V(\mathfrak{j}_I^{\mathrm{mov}}) = \bigcup_{I \not\subseteq gHg^{-1}} U_g, \\ \mathcal{P}_I^{\mathrm{mov}} &:= H^1_{\mathfrak{j}_I^{\mathrm{mov}}}(A_N), \qquad\qquad \mathcal{P}_I^{\mathrm{ram}} := H^0_{\mathfrak{d}_N}(\mathcal{P}_I^{\mathrm{mov}}), \\ \mathfrak{d}_N &:= \operatorname{Fitt}_0^{A_N} \Omega_{A_N/B}. \end{aligned}$$

Thus $\mathfrak{d}_N$ is the Kähler different; at height one its vanishing detects the non-étale locus. Because A_N is a domain and U_I^{mov} is dense, $H^0_{\mathfrak{j}_I^{\mathrm{mov}}}(A_N) = 0$. The spectral sequence for successive supports, using $\Gamma_{\mathfrak{d}_N}\Gamma_{\mathfrak{j}_I^{\mathrm{mov}}} = \Gamma_{\mathfrak{d}_N + \mathfrak{j}_I^{\mathrm{mov}}}$, therefore gives

$$\mathcal{P}_I^{\mathrm{ram}} \cong H^1_{\mathfrak{d}_N + \mathfrak{j}_I^{\mathrm{mov}}}(A_N).$$

If E is a hidden ramified divisor with $I = I_E$, then the pulled-back boundary dictionary and the DVR calculation give

$$(\mathcal{P}_I^{\mathrm{ram}})_{\mathfrak{p}_E} \cong \operatorname{Frac}((A_N)_{\mathfrak{p}_E})/(A_N)_{\mathfrak{p}_E},$$

whose pole orders are unbounded; hence $\mathcal{P}_I^{\mathrm{ram}}$ is not finite. This is the nonfiniteness half of a prospective contradiction.

There is nevertheless a canonical finite bounded-pole lattice. Write

$$T^\dagger := \{z \in N : \operatorname{Tr}_{N/K}(zT) \subseteq B\} \cong \operatorname{Hom}_B(T, B), \qquad T := A_N.$$

Finite flatness and generic separability identify $T^\dagger$ with the finite B-module $\operatorname{Hom}_B(T, B)$ through the trace pairing. Because B is normal, the trace of every element integral over B lies in B, so $T \subseteq T^\dagger$. Consequently

$$Q_{\mathrm{tr}} := T^\dagger / T$$

is finite. If $\mathfrak{q}_E = \mathfrak{p}_E \cap B$, then after localizing the DVR extension over $B_{\mathfrak{q}_E}$, separating its unramified part, and choosing a uniformizer π_E of $T_{\mathfrak{p}_E}$, the characteristic-zero tame formula gives the bounded codifferent stage

$$(Q_{\mathrm{tr}})_{\mathfrak{p}_E} \cong \pi_E^{-(e_E - 1)} T_{\mathfrak{p}_E} / T_{\mathfrak{p}_E}.$$

Here "codifferent" means the trace-dual lattice. Its inverse agrees with the Kähler different after height-one localization; no global invertibility or global identification with $\mathrm{Fitt}_0\,\Omega_{T/B}$ is used [The26c, The26d].

Write $\omega_Z := \mathrm{Hom}_B(T, \omega_Y)$ for the finite-duality canonical module. Trivializing ω_Y by $dP \wedge dQ$ identifies ω_Z with $T^\dagger \cong \mathrm{Hom}_B(T, B)$ [The26e]. If

$$\eta_F := c^{-1} dP \wedge dQ,$$

then the Keller identity restricts η_F to $d(gx_1) \wedge d(gx_2)$ on the sheet X_g. Under the trace identification,

$$T\eta_F \subseteq \omega_Z, \qquad Q_F^{\mathrm{sec}} := \omega_Z / T\eta_F \cong T^\dagger / T = Q_{\mathrm{tr}}.$$

Thus the "secant canonical-form target" is not a second finite module: it is the same codifferent quotient described geometrically. At a tame ramified divisor one has $\eta_F = u\pi_E^{e_E - 1}\omega_E$ for a unit u and a local generator ω_E of ω_Z, recovering the bounded-pole formula above. The Hilbert–Burch trivialization of Proposition 3.7 supplies a natural generic-sheet presentation, but identifying its chosen generator with η_F is a separate comparison and is not needed for the finite target itself.

The natural inclusion runs from this bounded stage into principal parts, not in the reverse direction:

$$(Q_{\mathrm{tr}})_{\mathfrak{p}_E} \hookrightarrow \mathrm{Frac}(T_{\mathfrak{p}_E}) / T_{\mathfrak{p}_E}.$$

Indeed, every $T_{\mathfrak{p}_E}$-linear map from the divisible module $\mathrm{Frac}(T_{\mathfrak{p}_E})/T_{\mathfrak{p}_E}$ to a finite $T_{\mathfrak{p}_E}$-module is zero. Thus finite flatness does not supply an injection of the entire ramification-supported principal-parts module into Q_{tr}; such an injection would already exclude a hidden ramified divisor.

This bounded stage nevertheless gives a canonical diagnostic. The open-complement exact sequence embeds $\mathcal{P}_I^{\mathrm{mov}} \cong \Gamma(U_I^{\mathrm{mov}}, \mathcal{O}_Z)/T$ into N/T; hence it embeds $\mathcal{P}_I^{\mathrm{ram}}$, while $T \subseteq T^\dagger \subseteq N$ embeds Q_{tr} in the same module. Define the trace-comparison ideal

$$\mathfrak{c}_I^{\mathrm{tr}} := \{s \in T : s\mathcal{P}_I^{\mathrm{ram}} \subseteq Q_{\mathrm{tr}}\}.$$

It is important to impose this condition only on the ramification-supported classes. Imposing it on the whole ring $\Gamma(U_I^{\mathrm{mov}}, \mathcal{O}_Z)$ would also detect unrelated unramified boundary divisors and can force the ideal to be zero for reasons stronger than the hidden-inertia problem.

Multiplication defines a canonical map

$$\Phi_I : \mathcal{P}_I^{\mathrm{ram}} \longrightarrow \mathrm{Hom}_T(\mathfrak{c}_I^{\mathrm{tr}}, Q_{\mathrm{tr}}), \qquad \Phi_I(\xi)(s) = s\xi.$$

The target is finite: T is Noetherian, so $\mathfrak{c}_I^{\mathrm{tr}}$ is finite, and both Q_{tr} and the displayed Hom module are finite. More sharply, if E is hidden with $I_E = I$, then

$$\mathfrak{c}_I^{\mathrm{tr}} = 0.$$

Indeed, let $0 \neq s \in \mathfrak{c}_I^{\mathrm{tr}}$ and localize at $\mathfrak{p}_E$. Multiplication by the nonzero element s on the divisible module $\mathrm{Frac}(T_{\mathfrak{p}_E})/T_{\mathfrak{p}_E}$ is surjective. Its image therefore still has arbitrarily deep poles and cannot lie in the bounded finite module $(Q_{\mathrm{tr}})_{\mathfrak{p}_E}$, a contradiction. Thus the exact prospective target is simply

$$\mathfrak{c}_I^{\mathrm{tr}} \neq 0 \qquad \text{for every inertia subgroup } I = I_E \text{ occurring at a ramified divisor } E,$$

or, more strongly, for every nontrivial $I \le G$. More concretely, one may construct a nonzero secant/frame ideal $0 \neq \mathfrak{s}_I^{\mathrm{sec}} \subseteq \mathfrak{c}_I^{\mathrm{tr}}$. This transporter construction packages the desired uniform pole bound; trace duality alone gives no reason for the transporter to be nonzero.

The fact that A_N or a boundary quotient is finite over its base does not give the opposite conclusion. For example, let

$$B_0 = \mathbb{C}[a,t] \subseteq R = \mathbb{C}[s,t], \qquad a = s^e.$$

Then R is finite free over B_0, and the boundary curve $R/(s) \cong B_0/(a)$ is finite over its image, whereas

$$H^1_{(s)}(R) = R_s/R = \bigcup_{m\ge 1} s^{-m}R/R$$

is not finitely generated. What is missing is a uniform transverse pole bound. The precise target is a Keller-specific theorem showing that the relevant boundary classes lie in one fixed fractional codifferent stage

$$(T :_N \mathfrak{d}_N^m)/T := \{z \in N : \mathfrak{d}_N^m z \subseteq T\}/T,$$

or equivalently that they admit a uniform different-power annihilator; neither assertion follows from finite flatness and neither is constructed here. More explicitly, if some power of $\mathfrak{d}_N + \mathfrak{j}_I^{\mathrm{mov}}$ uniformly annihilated the displayed local-cohomology module for every relevant I, local-cohomology finiteness (using the degree-zero vanishing above) and the DVR detector would force its vanishing and hence coverage [The26k, The26g]. Likewise, the secant determinant detects marked versus moving generic sheets, but it vanishes on a moving sheet and by itself gives no nonzero parameter that bounds boundary poles.

The inverse-Jacobian frame supplies plausible input for this last detection problem. Writing $c = \det JF$, the polynomial derivations

$$\partial_P = c^{-1}\left(Q_{x_2}\frac{\partial}{\partial x_1} - Q_{x_1}\frac{\partial}{\partial x_2}\right), \qquad \partial_Q = c^{-1}\left(-P_{x_2}\frac{\partial}{\partial x_1} + P_{x_1}\frac{\partial}{\partial x_2}\right)$$

restrict to the coordinate derivations on $K = \mathbb{C}(P,Q)$. Their unique extensions to the finite separable normal closure commute with G, hence preserve every conjugate affine ring gA. In the standard tame height-one model after a suitable completed Kummer base change (including a root of the unit), $u = t^e$, and differentiating gives $\partial t/\partial u = (et^{e-1})^{-1}$, the expected inverse-different denominator. Intrinsically, only its valuation $-(e-1)$ is used. A prospective secant–trace landing theorem would use the evaluated full secant matrix or its adjugate, together with this frame, to produce a nonzero $\mathfrak{s}_I^{\mathrm{sec}}$. Neither trace landing nor nonvanishing of such an ideal is asserted here.

For clarity, the logical status of the constructions used above is:

Construction	Role	Status
Sep_2, Cov_2, and PRR	Ideal, valuation, and differential descriptions of absence of divisorial ramification	Equivalent for each planar Keller map; universal validity is equivalent to $JC(2)$
PBC	Global boundary/properness description	Equivalent to automorphy for each map; not known universally
Torsion of $[R_{\bar{\nu}}]$	Sufficient divisor-class mechanism	Established for normal/Galois L/K, hence for generic degree two
Finiteness or uniform annihilation of $\mathcal{P}_I^{\mathrm{ram}}$	Sufficient principal-parts mechanism	Hidden inertia gives the opposite nonfiniteness; finite control is not derived from the Keller data
Secant–trace/codifferent boundedness	Possible source of a fixed bounded pole stage	Secant coevaluation and the trace-dual/canonical-form target are explicit; no secant/frame theorem supplies a nonzero multiplier landing all relevant classes in it

Thus the purity argument is the unique endgame. The divisor-class and principal-parts constructions are possible ways of establishing its height-one premise, not parallel conclusions that the reader must combine.

Lean correspondence. The formal planar implication chains are collected in `CollisionIdeals/Planar.lean`. The development takes the following established results as explicit inputs: planar Ax–Grothendieck, branch purity, and finite-étale rigidity of $\mathbb{A}^2_{\mathbb{C}}$. It also takes a supplied `PlanarKellerCollisionModelF`. The positive `PlanarHiddenInertia` predicate and the implication from `PlanarConjugateCoverage` to `PlanarNoHiddenInertia` are formalized; constructing the model and proving coverage remain separate formalization tasks. The fixed-locus ideal `fixedLocusIdeal`, the moving-boundary sum `movingBoundaryIdeal`, and the height-one condition `BoundaryIdealData.PlanarBoundarySeparation` are formalized in `CollisionIdeals/Planar/BoundarySeparation.lean`. The Lean predicate quantifies over all nontrivial subgroups; the manuscript's prime-order formulation is its equivalent Cauchy reduction. The theorem `BoundaryIdealData.ramifiedCodimensionOnePoint_isEmpty` connects that condition to the existing no-ramification endgame. The pulled-back open `pulledBackConjugateAffineOpen`, its closed complement `pulledBackConjugateBoundary`, and its vanishing ideal `pulledBackConjugateBoundaryIdeal` are constructed canonically; `pulledBackConjugateBoundaryIdeal_le_ramifiedPrime` formalizes the valuation-theoretic containment at a ramified divisor. The exact pointwise coverage assertion and the universal claims $\mathrm{PBC}(F)$ or $\mathrm{PRR}(F)$ remain hypotheses at the present Lean boundary, not conclusions of the formalization. The local-cohomology target is connected to the downstream endgame by the explicit input `BoundaryCoherenceBridge` and theorem `planarVanishing_of_boundaryCoherence`; the bridge itself is not proved. Likewise, `RamificationRigidityBridge` isolates the unformalized finite-length support step in the divisorial route. The stable local-cohomology object `BoundaryPrincipalParts` is defined in `CollisionIdeals/BoundaryPrincipalParts.lean`. The ramification-supported objects `RamificationSupportedBoundaryPrincipalParts` and `HasUniformCombinedSupportAnnihilator` are kept in the prospective module `CollisionIdeals/Planar/Research/PrincipalPartsStrategy.lean`. The abstract nonfiniteness mechanism for unbounded locally power-torsion principal parts is proved in `CollisionIdeals/Planar/Research/UnboundedPrincipalParts.lean`; finite comparison control and the resulting direct contradiction are formalized in `CollisionIdeals/Planar/Research/FinitePrincipalPartsControl.lean`. The evaluation of the planar secant relation on an arbitrary pair of collision sheets, including its vanishing when a coordinate is moved, is formalized in `CollisionIdeals/Planar/ConjugateSecantEvaluation.lean`. The prospective modules are collected under the opt-in umbrella `CollisionIdeals/Planar/Research.lean`, rather than the stable `CollisionIdeals/Planar.lean` import spine. The generic trace-integral submodule `TraceIntegralSubmodule`, the transporter `boundedStageComparisonIdeal`, and its canonical multiplication map `boundedStageComparisonMap` are also formalized abstractly in the research layer. Their specialization to the finite normalization—including the trace-dual/Hom identification, the open-complement embeddings, the height-one DVR calculation, and especially a nonzero secant multiplier— remains to be constructed. The boundary-coherence implication and the finite-length differentials criterion likewise remain separate formalization tasks. The properness and divisor-class criteria above are likewise not presently part of the Lean development.

Remark (Exact remaining content of classical $JC(2)$)*.* The traditional complex conjecture $JC(2)$ has no boundary, coverage, or module-finiteness condition in its definition: it asserts simply that every planar Keller map is a polynomial automorphism. Corollary 3.32 proves that its universal content is equivalent to the boundary formulations above. Theorem 3.14 settles every normal marked extension, and Corollary 3.15 settles generic degree at most two. Consequently, to prove $JC(2)$ it remains to show that every planar Keller map with L/K nonnormal and $d_F \geq 3$ satisfies

$$\operatorname{codim}_Z V(\mathfrak{a}_C + \mathfrak{j}_C^{\mathrm{mov}}) \geq 2 \quad \text{for every prime-order } C \leq \operatorname{Gal}(N/K).$$

Equivalently, one must exclude a common height-one fixed–moving boundary component in the normal closure of every remaining nonnormal extension. This is an exact reformulation of the unresolved part of $JC(2)$, not an additional hypothesis silently built into the classical conjecture.

4 The cubic S_3 obstruction in dimension three

We now specialize the dimension-independent constructions of Section 2 to dimension three and isolate the separable nonnormal cubic case.

Let $S_3 = \operatorname{Perm}(\{1,2,3\})$ be the symmetric group on three letters.

Proposition 4.1 (Dimension-three counterexample criterion)**.** *For a polynomial map*

$$F\colon \mathbb{A}^3_{\mathbb{C}} \longrightarrow \mathbb{A}^3_{\mathbb{C}},$$

consider the following statements:

(1) $\det JF \in \mathbb{C}^\times$*;*

(2) *F is not a polynomial automorphism;*

(3) $R_F^\circ \neq \varnothing$*.*

Statements (2) *and* (3) *are equivalent. Consequently, F is a dimension-three Jacobian-conjecture counterexample exactly when* (1) *and* (3) *hold.*

Proof. By Corollary 2.4 and Theorem 2.15,

$$R_F^\circ = \varnothing \quad \Longleftrightarrow \quad I_R = I_\Delta \quad \Longleftrightarrow \quad F \in \operatorname{Aut}_{\mathrm{poly}}(\mathbb{A}^3_{\mathbb{C}}).$$

Negating this equivalence proves (2)$\Longleftrightarrow$(3). Combining this with (1) gives the final assertion. ◻

4.1 The cubic extension and its normal closure

Let

$$F = (F_1, F_2, F_3) \colon \mathbb{A}^3_{\mathbb{C}} \longrightarrow \mathbb{A}^3_{\mathbb{C}}$$

be dominant and generically finite. Let

$$A = \mathbb{C}[x_1, x_2, x_3], \qquad B = \mathbb{C}[F_1, F_2, F_3] \subseteq A, \qquad K = \mathrm{Frac}(B), \qquad L = \mathrm{Frac}(A),$$

and let

$$\theta \colon K \otimes_B A \longrightarrow L, \qquad \lambda \otimes a \longmapsto \lambda a,$$

be the isomorphism of Theorem 2.6. Suppose that L/K is separable and nonnormal, and choose a power basis with generator α and degree three:

$$L = K(\alpha), \qquad [L : K] = 3.$$

If $m_\alpha \in K[T]$ is the minimal polynomial of α, let $h_\alpha \in L[T]$ be determined by

$$m_\alpha(T) = (T - \alpha)h_\alpha(T) \qquad \text{in } L[T].$$

Let N/K be the normal closure of L/K, containing L through the marked inclusion. Let

$$G = \mathrm{Gal}(N/K), \qquad H = \mathrm{Gal}(N/L).$$

Recall from Section 2.5 that

$$G/H \xrightarrow{\sim} \mathrm{Hom}_K(L, N), \qquad gH \longmapsto g|_L, \qquad \mathrm{core}_G(H) = 1.$$

The final equality follows because N is the normal closure of L/K; equivalently, the action of G on the conjugate embeddings of L is faithful.

For $Y = \operatorname{Spec} B$, the finite-normalization construction of Section 2.5 gives

$$Z = \mathrm{Norm}_N(Y) \longrightarrow \overline{X} = \mathrm{Norm}_L(Y) \longrightarrow Y.$$

If $E \in Z^{(1)}$, with decomposition group D_E and inertia group I_E, then its conjugate centers are indexed by $\mathcal{Q}_E = D_E \backslash G/H$. For $\omega = D_E gH$, Proposition 2.11 states

$$e\big(\mathrm{cent}(E, \omega)/\nu(E)\big) = i_E(\omega) = [I_E : I_E \cap gHg^{-1}].$$

The following proposition is the standard residual-factor description of a separable nonnormal cubic extension. We include its short proof to fix the marked identification with N.

Proposition 4.2 (Cubic residual-factor identification)**.** *The polynomial h_α is irreducible over L, and there is an L-algebra isomorphism*

$$\varphi \colon L[T]/(h_\alpha) \xrightarrow{\sim}_L N.$$

Consequently,

$$\Psi \colon L \otimes_K L \xrightarrow{\sim}_L L \times N, \qquad \mathrm{pr}_1 \circ \Psi = \mu_L.$$

Proof. Write
$$m_\alpha(T) = T^3 + a_2T^2 + a_1T + a_0.$$
If the quadratic h_α were reducible over L, it would have a root $\beta \in L$. Separability gives $\beta \neq \alpha$, and the third root would be
$$\gamma = -a_2 - \alpha - \beta \in L.$$
Thus m_α would split over L. Since $L = K(\alpha)$, the extension L/K would then be the splitting field of the separable polynomial m_α, hence normal. This contradicts the assumed nonnormality, so h_α is irreducible.

Let β denote the image of T in $L[T]/(h_\alpha)$. This field contains the three roots
$$\alpha, \qquad \beta, \qquad -a_2 - \alpha - \beta$$
of m_α, and is generated over L by β. It is therefore the splitting field of m_α, which is the normal closure N of L/K. Choose the resulting L-algebra isomorphism
$$\varphi\colon L[T]/(h_\alpha) \xrightarrow{\sim}_L N.$$

Let
$$\Phi_\alpha\colon L \otimes_K L \xrightarrow{\sim}_L L \times L[T]/(h_\alpha)$$
be the marked-root decomposition of Theorem 2.7. It satisfies
$$\mathrm{pr}_1 \circ \Phi_\alpha = \mu_L.$$
Composing its second factor with φ, let
$$\Psi = (\mathrm{id}_L \times \varphi) \circ \Phi_\alpha.$$
Since $\mathrm{id}_L \times \varphi$ leaves the first factor unchanged, $\mathrm{pr}_1 \circ \Psi = \mu_L$. □

Theorem 4.3 (Galois group of a nonnormal cubic extension)**.**
$$\mathrm{Gal}(N/K) \cong S_3.$$

Proof. Proposition 4.2 identifies N with the degree-two extension $L[T]/(h_\alpha)$ of L. Hence
$$|H| = [N : L] = 2.$$
The normal-closure identities recalled above give
$$\mathrm{core}_G(H) = 1, \qquad [G : H] = [L : K] = 3.$$
Hence the action of G on the three cosets of H gives
$$\rho\colon G \longrightarrow \mathrm{Perm}(G/H) \cong S_3, \qquad \ker \rho = \mathrm{core}_G(H) = 1.$$
Core-freeness therefore makes ρ injective, so
$$|G| \leq |S_3| = 6.$$
The index formula gives
$$|G| = [G : H]|H| = 3 \cdot 2 = 6.$$
Thus $|G| = 6$, and the injective homomorphism ρ is an isomorphism $G \cong S_3$. □

4.2 The cubic off-diagonal factor

Retain the cubic data of Section 4.1, and suppose in addition that F is Keller. Thus

$$B = \mathbb{C}[F_1, F_2, F_3] \subseteq A = \mathbb{C}[x_1, x_2, x_3], \qquad K = \operatorname{Frac}(B) \subseteq L = \operatorname{Frac}(A) = K(\alpha) \subseteq N,$$

where L/K is separable, nonnormal, and of degree three, and N/K is its normal closure. Let

$$\theta\colon K \otimes_B A \xrightarrow{\sim} L, \qquad \lambda \otimes a \longmapsto \lambda a.$$

Corollary 4.4 (Generic cubic S_3 collision)**.** *There is a K-algebra isomorphism*

$$\Psi_F\colon K \otimes_B C_F \xrightarrow{\sim}_K L \times N, \qquad \mathrm{pr}_1 \circ \Psi_F = \theta \circ (1 \otimes \bar{\mu}_F).$$

It satisfies

$$\Psi_F\big(\ker(\theta \circ (1 \otimes \bar{\mu}_F))\big) = 0 \times N, \qquad \operatorname{Gal}(N/K) \cong S_3.$$

Consequently,

$$\operatorname{Obs}(F) \neq 0, \qquad I_R \subsetneq I_\Delta, \qquad R_F^\circ \neq \varnothing, \qquad F \notin \operatorname{Aut}_{\mathrm{poly}}(\mathbb{A}^3_{\mathbb{C}}).$$

Proof. Theorem 2.6 gives a K-algebra isomorphism

$$\Phi_F\colon K \otimes_B C_F \xrightarrow{\sim}_K L \otimes_K L$$

such that

$$\mu_L \circ \Phi_F = \theta \circ (1 \otimes \bar{\mu}_F).$$

Proposition 4.2 gives

$$\Psi\colon L \otimes_K L \xrightarrow{\sim}_L L \times N, \qquad \mathrm{pr}_1 \circ \Psi = \mu_L.$$

After restricting Ψ from L to K, let

$$\Psi_F = \Psi \circ \Phi_F.$$

It follows that

$$\mathrm{pr}_1 \circ \Psi_F = \theta \circ (1 \otimes \bar{\mu}_F), \qquad \Psi_F\big(\ker(\theta \circ (1 \otimes \bar{\mu}_F))\big) = 0 \times N \neq 0.$$

Suppose that the diagonal-evaluation homomorphism $\bar{\mu}_F\colon C_F \to A$ were injective. Since K is a localization of B, it is flat over B, so $1 \otimes \bar{\mu}_F$ would also be injective. Theorem 2.6 shows that θ is an isomorphism. Therefore $\theta \circ (1 \otimes \bar{\mu}_F)$ would be injective, contradicting the nonzero kernel $0 \times N$ above. Hence $\ker(\bar{\mu}_F) \neq 0$, and Proposition 2.2 gives $\operatorname{Obs}(F) \neq 0$. Proposition 2.1 shows that $I_R \subseteq I_\Delta$ strict, and Corollary 2.4 gives $R_F^\circ \neq \varnothing$.

Finally, Theorem 4.3 identifies $\operatorname{Gal}(N/K)$ with S_3. If F were a polynomial automorphism, Theorem 2.15 would give $\operatorname{Obs}(F) = 0$, contrary to $\operatorname{Obs}(F) \neq 0$. Since F is Keller, it is a dimension-three Jacobian-conjecture counterexample. $\square$

For the same cubic map and field tower, let

$$X = \operatorname{Spec} A, \qquad Y = \operatorname{Spec} B, \qquad G = \operatorname{Gal}(N/K), \qquad H = \operatorname{Gal}(N/L).$$

Suppose there is a diagram over Y

$$\mathcal{D} = \Big(Z = \operatorname{Norm}_N(Y) \xrightarrow{\pi} \overline{X} = \operatorname{Norm}_L(Y) \xrightarrow{\bar{\nu}} Y,\ j\colon X \hookrightarrow \overline{X}\Big)$$

in which π and $\bar{\nu}$ are finite, j is an open immersion, $\bar{\nu} \circ j = f_F$, and $f_F\colon X \to Y$ is étale. Let

$$\partial X = \overline{X} \setminus j(X), \qquad G \cong S_3$$

by Theorem 4.3.

Corollary 4.5 (Hidden inertia in the S_3-normalization)**.**

$$\mathcal{H}(\mathcal{D}) = \mathrm{Ram}^{(1)}(Z/Y).$$

Proof. Apply Proposition 2.14 to the displayed normalization diagram. □

Remark (Ordered roots)*.* Let

$$m_\alpha(T) = T^3 + a_2T^2 + a_1T + a_0$$

have roots $\alpha_1, \alpha_2, \alpha_3$ in N. For distinct i, j, k,

$$\alpha_k = -a_2 - \alpha_i - \alpha_j, \qquad K(\alpha_i, \alpha_j) = K(\alpha_1, \alpha_2, \alpha_3) = N.$$

Thus the off-diagonal factor, which marks two distinct roots, is already the ordered-root normal closure.

Lean correspondence. `CollisionIdeals/ComplexThree/CubicGaloisGroup.lean`, `CollisionIdeals/ComplexThree/S3Collision.lean`, `CollisionIdeals/ComplexThree/OffDiagonal.lean`, `CollisionIdeals/ComplexThree/CubicS3.lean`, and `CollisionIdeals/ComplexThree/JacobianConjecture.lean`. The Lean cubic witness takes the residual-field isomorphism and $H \neq 1$ as explicit inputs; Proposition 4.2 derives them here from nonnormality. Hidden inertia is formalized pointwise at a supplied ramified divisor.

Remark (Exact relation with classical complex $JC(3)$)*.* The traditional complex three-dimensional Jacobian conjecture asserts that every polynomial map

$$F\colon \mathbb{A}^3_{\mathbb{C}} \longrightarrow \mathbb{A}^3_{\mathbb{C}}, \qquad \det JF \in \mathbb{C}^\times,$$

is a polynomial automorphism; equivalently, every such map satisfies $\mathrm{Obs}(F) = 0$. Corollary 4.4 proves the following implication under a prescribed function-field hypothesis:

$$\begin{aligned} &\det JF \in \mathbb{C}^\times, \qquad [L:K] = 3, \qquad L/K \text{ nonnormal} \\ \Longrightarrow\quad &\mathrm{Obs}(F) \neq 0 \quad\Longrightarrow\quad F \text{ is a counterexample to } JC(3). \end{aligned}$$

Consequently, the existence of one such Keller map would disprove $JC(3)$, while $JC(3)$ implies that no such map exists. This section does not construct such a map, and it does not assert that every possible counterexample to $JC(3)$ has generic degree three. It identifies the generic collision geometry of this cubic nonnormal subclass.

5 Conclusion

The obstruction ideal $\mathrm{Obs}(F)$ vanishes exactly when the scheme-theoretic collision relation $X \times_Y X$ equals the diagonal. In dimension two there is one endgame with several exact formulations. PlanarBoundaryCoherence expresses it through disappearance of the whole deleted boundary. PlanarRamificationRigidity, moving-sheet coverage, and boundary separation express its height-one premise in differential, valuation-theoretic, and finite-group language. Only after that premise is established do purity and finite-étale rigidity make the normal closure trivial and force $\mathrm{Obs}(F) = 0$. The image of the deleted boundary is exactly the nonproperness set, and the boundary divisor-class argument proves the normal/Galois case unconditionally. Universal validity of the equivalent planar criteria is exactly $JC(2)$, not a result asserted here. In the cubic dimension-three class, by contrast, the nonzero generic factor is the ordered-root S_3-normal closure. Both statements arise from the same collision, off-diagonal, and conjugate-sheet constructions; their point is to locate the obstruction, respectively, at the normalization boundary and on the generic off-diagonal sheet.

Under PlanarBoundaryCoherence or PlanarRamificationRigidity in dimension two, under the conditions of Section 4 in dimension three, and with $\mathrm{Ram}^{(1)}(Z/Y) \neq \varnothing$ in the cubic normalization, the contrast is

planar rigidity criteria	nonnormal cubic class in dimension three
$\mathrm{PBC}(F)$ or $\mathrm{PRR}(F)$	hidden inertia in the S_3-cover realized
$\mathrm{Obs}(F) = 0$	$\mathrm{Obs}(F) \neq 0$
$I_R = I_\Delta$	$I_R \subsetneq I_\Delta$
F is a polynomial automorphism	F is not a polynomial automorphism.

Motivating example. The explicit three-dimensional polynomial counterexample announced by Levent Alpöge [Alp26] and independently formalized in Isabelle/HOL by Ramos, Hulak, and de Queiroz [RHdQ26] motivated the cubic S_3 comparison. The formal verification establishes a constant nonzero Jacobian determinant and distinct rational points with the same image. That particular map is not used in the arguments above: the dimension-three result is the structural implication from the stated cubic and marked-normal-closure conditions.

The formalization uses Lean 4 and Mathlib [dMU21, The20].

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